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Gravitational Quasinormal Modes of Schwarzschild and Kerr Black Holes: Numerical Spectra and Their Geometric Interpretation

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Declaration

I hereby certify that this Dissertation, which is approximately 15 thousand words in length, has been written by me at the School of Physical and Chemical Sciences, Queen Mary University of London, that all material in this dissertation which is not my own work has been properly acknowledged, and that it has not been submitted in any previous application for a higher degree.

The sections which contain the report on the independent research work component of the project are Sections 5.2–5.3, 5.5–5.6, 6.1–6.5, 7.1–7.6 and 8.1–8.4.

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Abstract

A perturbed black hole relaxes by emitting gravitational waves whose late-time form is a sum of damped sinusoids, the quasinormal modes, with complex frequencies that depend only on the mass and spin of the hole. This dissertation develops the theory of these modes from the Schwarzschild and Kerr geometries and tests its main predictions numerically. The Regge–Wheeler equation is derived from the linearised Einstein equations, the quasinormal modes are defined through their boundary conditions, and the Schwarzschild spectrum is computed with an independent implementation of Leaver's continued-fraction method with the Nollert remainder, up to overtone number 130. The highly damped modes approach $\mathrm{Re}(M\omega) = \ln 3/8\pi$ from above with the expected $n^{-1/2}$ correction. A time-domain evolution of the same equation recovers the fundamental frequency to better than 0.03 per cent and reproduces the late-time power-law tails, with exponents $-(2\ell+3)$ for generic and $-(2\ell+4)$ for time-symmetric initial data. For Kerr black holes the fundamental modes with $\ell = |m| \leq 10$ and the overtones $n \leq 7$ of $\ell = m = 2$ are followed from $a/M = 0$ to 0.998 with the QNM package, and the co-rotating fundamental modes to $a/M = 0.99999$. At large ℓ the real part follows the orbital frequency of the prograde or retrograde equatorial photon orbit to 0.2 per cent. Near extremality the damping of the co-rotating fundamental modes approaches $\kappa/2$, where κ is the surface gravity, and the overtones form an evenly spaced ladder with spacing $0.978\,\kappa$ at $a/M = 0.998$. The $n = 5$ overtone of $\ell = m = 2$ leaves this ladder after a close approach to its neighbour near $a/M = 0.9$, so that the overtone label defined at zero spin does not follow the order of damping at high spin.

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1. Introduction

1.1. Black holes as probes of gravity

Black holes are the simplest macroscopic objects that general relativity predicts, and over the last decade they have become objects of direct observation. The first detection of gravitational waves from the merger of two black holes (Abbott et al. 2016) opened a channel in which the strong-field regime of gravity can be measured rather than inferred, and very long baseline interferometry now resolves the neighbourhood of the horizon of supermassive black holes. The lecture notes of Cardoso et al. (2025), which form the starting point of this project, organise the subject around three questions: whether black holes exist as the exact solutions that general relativity describes, whether cosmic censorship holds in the processes that could threaten it, and how astrophysical environments modify the gravitational-wave signals that black holes produce. The first of these questions is the one addressed here.

The gravitational-wave signal from the coalescence of two compact objects is conventionally divided into three stages. During the inspiral the two bodies orbit each other and slowly spiral together, and the signal can be modelled by expanding in the orbital velocity. At the merger the two horizons join and the dynamics is fully nonlinear. What remains afterwards is a single, distorted black hole. It returns to equilibrium by radiating away its deformation, and after a short transient this relaxation is dominated by a superposition of exponentially damped sinusoids,

$$h(t) \;\propto\; \sum_{\ell mn} A_{\ell mn}\, e^{-\mathrm{i}\omega_{\ell mn} t}, \qquad \mathrm{Im}\,\omega_{\ell mn} < 0, \tag{1.1}$$

where the $\omega_{\ell mn}$ are complex numbers. This last stage is the ringdown, and the complex frequencies are the quasinormal modes (QNMs) of the black hole.

1.2. Why quasinormal modes matter

The reason that the ringdown is of interest, beyond being the last part of a detectable signal, is the no-hair property of black holes. In general relativity the exterior of an isolated, stationary black hole in vacuum is described by the Kerr metric, which is fixed by two numbers, the mass M and the angular momentum $J = aM$ (Kerr 1963; Misner et al. 1973). The frequencies $\omega_{\ell mn}$ are therefore functions of M and a alone, and not of how the black hole was formed. If one mode is measured, its real and imaginary parts determine M and a. If a second mode is measured, its frequency is a prediction that can be checked, and a disagreement would mean that the remnant is not a Kerr black hole. This is the idea behind black-hole spectroscopy, in analogy with the use of atomic spectra to identify elements.

Each mode carries two independent numbers, an oscillation frequency $f = \mathrm{Re}\,\omega/2\pi$ and a damping time $\tau = 1/|\mathrm{Im}\,\omega|$. Their combination, the quality factor

$$Q = \frac{\mathrm{Re}\,\omega}{2\,|\mathrm{Im}\,\omega|} = \pi f \tau, \tag{1.2}$$

counts the number of oscillations in which the amplitude falls by a factor $e^{-\pi}$, and so measures how sharply defined the mode is. The first merger signal observed, GW150914, left a remnant that was inferred to have a mass of about $62\,\mathrm{M}_\odot$ and a spin $a \simeq 0.67$ (Abbott et al. 2016), and its ringdown was later used to test the no-hair property (Isi et al. 2019). Which overtones are present with enough amplitude to be measured is still a matter of debate, and the answer depends on both the frequencies, which are computed in this dissertation, and the excitation amplitudes, which are not.

1.3. A short history of the subject

The mathematical problem was posed by Regge and Wheeler (1957), who studied the stability of the Schwarzschild solution against small perturbations and reduced the odd-parity gravitational perturbations to a single wave equation. The even-parity counterpart was found by Zerilli (1970). The first indication that perturbed black holes ring came from the scattering calculations of Vishveshwara (1970), in which a pulse sent at a Schwarzschild black hole returns as a damped oscillation, and Press (1971) described the resulting long trains of waves as the signature of a vibrating black hole. The frequencies themselves were first computed accurately by Chandrasekhar and Detweiler (1975), and for rotating black holes by Detweiler

(1980).

A geometrical interpretation of the modes was given early on. Goebel (1972) noticed that for large angular momentum the real part of the frequency is the orbital frequency of the unstable circular photon orbit, and Ferrari and Mashhoon (1984) showed that the modes can be understood as the bound states of the inverted potential barrier. This picture, which was later formulated in terms of the Lyapunov exponent of the orbit (Cardoso et al. 2009), is tested numerically in Chapters 6 and 7. On the numerical side, the WKB method of Schutz and Will (1985) gave an analytic estimate of the lowest modes, and the continued-fraction method of Leaver (1985), which reaches the very high overtones, became the standard tool. Nollert (1993) used it to follow the overtones to large order and found that the real part tends to a constant, a number that Hod (1998) identified as $\ln 3/8\pi$ and that Motl and Neitzke (2003) derived analytically. The review articles of Kokkotas and Schmidt (1999), Nollert (1999) and Berti et al. (2009) give the state of the subject at different stages, and Chandrasekhar (1983) remains the standard reference for the analytic structure.

1.4. Aims, scope and sources

This project began as a structured study of the material in the lecture notes of Cardoso et al. (2025), namely the geometry of the Schwarzschild and Kerr solutions and the motion of particles in them (their Part I), and perturbation theory, scattering and superradiance (Part II), and of the review by Kokkotas and Schmidt (1999). It then developed into a computational study of the spectrum itself. The aims of this dissertation are:

1. to assemble the geometric background needed to state the perturbation problem, and the null-geodesic quantities (light ring, Lyapunov exponent) that are later compared with the spectrum;

2. to derive the Regge–Wheeler equation from the linearised Einstein equations, rather than quote it, and to state precisely how the quasinormal modes are defined and why they form a discrete spectrum;

3. to compute the Schwarzschild spectrum with an independent implementation of Leaver's method, including very highly damped overtones, and with a time-domain evolution that does not rely on the frequency-domain boundary conditions, and to compare both with the analytic approximations in the literature;

4. to map the gravitational QNM spectrum of Kerr black holes up to $\ell = 10$ and near-extremal spin, and to test the physical pictures that are used to interpret it.

The scope is limited in three ways. Only gravitational perturbations of vacuum black holes are considered, at linear order. Only the frequencies are computed and not the excitation amplitudes, which depend on the initial data. And for Kerr black holes I use the existing QNM package (Stein 2019) for the frequencies, so that the contribution there is the design and analysis of the scans and not the solver.

Chapters 2, 3 and 4 are a review, written to make the argument self-contained, and results that are known are quoted with references. Where I derived or checked a statement myself, for example by symbolic computation, this is stated in the text. Every numerical value in Chapters 5, 6 and 7 was computed for this dissertation, and the scripts are listed in Appendix D.

1.5. Conventions and units

Throughout, $G = c = 1$ and the metric signature is $(-,+,+,+)$. Perturbations depend on time as $e^{-\mathrm{i}\omega t}$, so that a damped mode has $\mathrm{Im}\,\omega < 0$. Some of the literature, including the review by Kokkotas and Schmidt (1999), uses $e^{+\mathrm{i}\omega t}$, in which case the sign of the imaginary part for a damped mode is reversed; all frequencies quoted here are in the first convention, and each frequency is the one with $\mathrm{Re}\,\omega > 0$ of the pair ω, $-\omega^*$ (Section 4.2). Frequencies are given as the dimensionless product $M\omega$, and spins as the dimensionless ratio a/M, written a when the meaning is clear. A mode with $M\omega = x - \mathrm{i}y$ in a black hole of mass M has

$$f = \frac{x}{2\pi}\frac{c^3}{GM} = 32.31\,x\left(\frac{\mathrm{M}_\odot}{M}\right)\ \mathrm{kHz}, \qquad \tau = \frac{GM}{c^3}\frac{1}{y} = 4.925\,\frac{M}{\mathrm{M}_\odot}\,\frac{1}{y}\ \mu\mathrm{s}. \tag{1.3}$$

As an example, the fundamental mode $M\omega = 0.37367 - 0.08896\,\mathrm{i}$ of Chapter 6 corresponds in a $10\,\mathrm{M}_\odot$ black hole to a frequency of 1.21 kHz and a damping time of 0.55 ms.

1.6. Outline

Chapter 2 summarises the Schwarzschild and Kerr geometries, their null orbits and the energetics of rotation. Chapter 3 sets up black-hole perturbation theory: the master equations, scattering, superradiance and the Teukolsky equation. Chapter 4

defines the quasinormal modes, discusses their general properties and describes the structure that is expected in the Schwarzschild and Kerr spectra. Chapter 5 describes the numerical methods and their validation. The Schwarzschild and Kerr results are presented in Chapters 6 and 7. Chapter 8 discusses them, and Chapter 9 gives the conclusions and suggestions for further work. The derivation of the Regge–Wheeler equation, the derivation of the recurrence relation used in the continued fraction, selected program listings and supplementary data are given in the appendices.

2. Black-hole spacetimes and null orbits

This chapter collects the properties of the Schwarzschild and Kerr solutions that are needed later. Most of the material follows Part I of Cardoso et al. (2025) and the standard texts (Chandrasekhar 1983; Misner et al. 1973). Its purpose in this dissertation is twofold: to define the quantities that enter the perturbation equations, and to introduce the circular photon orbits, which turn out to control the real part of the quasinormal frequencies at large angular momentum.

2.1. The Schwarzschild solution

In geometrised units the Einstein equations read $G_{\mu\nu} = 8\pi T_{\mu\nu}$, and in vacuum they reduce to $R_{\mu\nu} = 0$. The unique spherically symmetric vacuum solution, which is automatically static, is the Schwarzschild metric

$$\mathrm{d}s^2 = -f\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right), \qquad f(r) = 1 - \frac{2M}{r}, \tag{2.1}$$

where M is the mass of the black hole. The metric has two special radii. The surface $r = 0$ is a genuine curvature singularity, as the Kretschmann scalar shows,

$$R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta} = \frac{48M^2}{r^6}, \tag{2.2}$$

which is a scalar and so takes the same value in every coordinate system. At $r = 2M$ the same scalar is finite, and the apparent singularity of the metric components is a failure of the coordinates. This is made explicit in Kruskal–Szekeres coordinates (T, X), defined by

$$X^2 - T^2 = \left(\frac{r}{2M} - 1\right)e^{r/2M}, \qquad \frac{T}{X} = \tanh\frac{t}{4M}, \tag{2.3}$$

in which the metric is $\mathrm{d}s^2 = (32M^3/r)\, e^{-r/2M}(-\mathrm{d}T^2 + \mathrm{d}X^2) + r^2\mathrm{d}\Omega^2$ and is regular across $r = 2M$. The surface $r = 2M$ is the event horizon. A freely falling observer crosses it in a finite proper time, whereas the coordinate time t of a distant observer diverges logarithmically.

Perturbations of the black hole are most conveniently described with the tortoise coordinate r_*, defined by $\mathrm{d}r_*/\mathrm{d}r = 1/f$,

$$r_* = r + 2M \ln\left(\frac{r}{2M} - 1\right). \tag{2.4}$$

It maps the exterior $2M < r < \infty$ onto the whole real line, with $r_* \to -\infty$ at the horizon and $r_* \to +\infty$ at infinity. The radial null rays satisfy $t \mp r_* = \text{const}$ and so move with unit speed in the (t, r_*) plane. The horizon is pushed to infinite distance in this coordinate, which is why the wave equations of the next chapter take the form of a one-dimensional wave equation on a line.

2.2. Uniqueness and the absence of hair

The statement that a black hole is described by very few parameters has a precise form in general relativity. The Kerr metric of Section 2.4 is the unique stationary, axisymmetric, asymptotically flat solution of the vacuum Einstein equations with a regular horizon, so that the exterior of an isolated black hole is described by M and a alone (Cardoso et al. 2025; Chandrasekhar 1983). The same conclusion holds in several cases when matter fields are added, which is the origin of the phrase that black holes have no hair. A simple example is a static scalar field Φ with $\Box\Phi = 0$ on the Schwarzschild background that is regular on the horizon and tends to a constant Φ_0 at infinity. Since $\nabla_\mu\left[(\Phi - \Phi_0)\nabla^\mu\Phi\right] = (\nabla\Phi)^2$ for such a field, integrating over the exterior gives

$$\int (\nabla\Phi)^2 \,\mathrm{d}V = \oint (\Phi - \Phi_0)\, \nabla^\mu\Phi \,\mathrm{d}\Sigma_\mu. \tag{2.5}$$

The surface term vanishes at the horizon, where Φ and Φ' are finite and $g^{rr} = f \to 0$, and at infinity, where $\Phi - \Phi_0 = O(1/r)$ and $\Phi' = O(1/r^2)$. For a static field $(\nabla\Phi)^2 = f\,\Phi'^2 \geq 0$, so that Φ' vanishes everywhere and $\Phi \equiv \Phi_0$. Cardoso et al. (2025) give the version of this argument for a scalar field with a convex potential. The relevance for what follows is that the quasinormal frequencies of a black hole in vacuum are fixed by M and a, which is what makes them useful as a test of the theory.

2.3. Motion of particles: the light ring

Free particles follow geodesics. In Schwarzschild the time-translation and rotational symmetries give two conserved quantities per particle, the energy $E = f\dot{t}$ and the angular momentum $L = r^2\dot{\phi}$ for motion in the equatorial plane, where a dot is the derivative with respect to an affine parameter. The normalisation $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -\delta$, with $\delta = 0$ for light and $\delta = 1$ for massive particles, then gives the radial equation

$$\dot{r}^2 = E^2 - V_{\rm eff}(r), \qquad V_{\rm eff} = f\left(\frac{L^2}{r^2} + \delta\right). \tag{2.6}$$

For light the effective potential $V_{\rm null} = fL^2/r^2$ has a single maximum, where $V'_{\rm null} = 0$. Writing $V_{\rm null} = L^2(r^{-2} - 2Mr^{-3})$, the derivative vanishes at

$$r_{\rm ph} = 3M, \qquad V_{\rm null}(r_{\rm ph}) = \frac{L^2}{27M^2}, \tag{2.7}$$

which is the circular photon orbit, or light ring. It is unstable, because it is a maximum of the effective potential. A photon coming from infinity with impact parameter $b = L/E$ is captured if E^2 exceeds the barrier, that is if $b < b_c = 3\sqrt{3}\,M$, so that the high-frequency absorption cross-section of the black hole is the geometrical one,

$$\sigma_{\rm abs} = \pi b_c^2 = 27\pi M^2. \tag{2.8}$$

The instability of the light ring is characterised by a Lyapunov exponent. For a small displacement δr from the orbit, equation (2.6) gives $\delta\ddot{r} = -\frac{1}{2}V''_{\rm null}\,\delta r$, and converting to coordinate time with $\dot{t} = E/f = 3E$ at $r = 3M$ gives $\delta r \propto e^{\lambda_{\rm L} t}$ with

$$\lambda_{\rm L}^2 = -\frac{V''_{\rm null}(r_{\rm ph})}{2\dot{t}^2} = \frac{2L^2/81M^4}{2\cdot 9E^2} = \frac{1}{27M^2}, \qquad \lambda_{\rm L} = \frac{1}{3\sqrt{3}\,M}, \tag{2.9}$$

where I used $L^2/E^2 = 27M^2$ for the critical orbit. The angular velocity of the orbit seen from infinity is $\Omega_c = {\rm d}\phi/{\rm d}t = fL/(r^2E)$, which at the light ring has the same value, $\Omega_c = 1/(3\sqrt{3}\,M)$. In Schwarzschild the orbital frequency and the Lyapunov exponent of the light ring are therefore equal. Both reappear in Chapter 6: the real and imaginary parts of the quasinormal frequencies at large ℓ are Ω_c and $\lambda_{\rm L}$ times simple factors.

For massive particles ($\delta = 1$) circular orbits exist only for $r \geq 3M$, and they are stable only for $r \geq 6M$. The innermost stable circular orbit at $r = 6M$ marks the inner edge of a thin accretion disc, but plays no role in the perturbation problem,

and I do not discuss it further.

2.4. The Kerr solution

Real black holes rotate, and the rotating generalisation of equation (2.1) is the Kerr solution. In Boyer–Lindquist coordinates the metric of a black hole with mass M and angular momentum $J = aM$ is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 + \Sigma\,\mathrm{d}\theta^2 \\ &+ \left(r^2 + a^2 + \frac{2Ma^2r\sin^2\theta}{\Sigma}\right)\sin^2\theta\,\mathrm{d}\phi^2, \end{aligned} \tag{2.10}$$

with $\Sigma = r^2 + a^2\cos^2\theta$ and $\Delta = r^2 - 2Mr + a^2$ (Kerr 1963). The horizons are the roots of $\Delta = 0$,

$$r_\pm = M \pm \sqrt{M^2 - a^2}, \tag{2.11}$$

so that a horizon exists only if $a \le M$; for $a > M$ the singularity would be naked. The cross term $g_{t\phi} = -2Mar\sin^2\theta/\Sigma$ drags every observer around the hole. Inside the ergoregion, which extends to $r_\mathrm{E}(\theta) = M + \sqrt{M^2 - a^2\cos^2\theta}$, the dragging is so strong that no observer can remain at rest with respect to infinity, and ∂_t is spacelike. The horizon rotates rigidly with angular velocity and has surface gravity

$$\Omega_\mathrm{H} = \frac{a}{r_+^2 + a^2} = \frac{a}{2Mr_+}, \qquad \kappa = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \tag{2.12}$$

and the Hawking temperature is $T_\mathrm{H} = \kappa/2\pi$. The surface gravity decreases monotonically with spin and vanishes in the extremal limit $a \to M$, where the two horizons coincide. This will matter for the damping of near-extremal modes in Section 7.4.

2.4.1. Energy extraction and the irreducible mass

The ergoregion allows rotational energy to be extracted. In the process proposed by Penrose (1969), a particle that enters the ergoregion splits into two, one of which falls into the hole with negative energy as measured from infinity and the other escapes with more energy than the original particle had. The extractable energy is limited by the area theorem, according to which the horizon area

$$A = 4\pi(r_+^2 + a^2) = 8\pi M^2\left(1 + \sqrt{1 - a^2/M^2}\right) \tag{2.13}$$

cannot decrease in classical processes. The area defines the irreducible mass $M_{\rm irr} = \sqrt{A/16\pi}$, which cannot be reduced either, and the mass is related to it by (Christodoulou 1970)

$$M^2 = M_{\rm irr}^2 + \frac{J^2}{4M_{\rm irr}^2}. \tag{2.14}$$

For an extremal black hole $M_{\rm irr} = M/\sqrt{2}$, so that at most $1 - 1/\sqrt{2} \approx 29$ per cent of the mass is rotational energy that could in principle be extracted. The wave analogue of the Penrose process, superradiant scattering, is discussed in Section 3.7, and it is through superradiance that rotation affects the quasinormal modes.

The bound $a \leq M$ is also protected dynamically. A test particle with angular momentum $L = Mh$ that is absorbed by a black hole of spin $j = a/M$ changes it according to $\mathrm{d}j = (h - 2j)\,\mathrm{d}M/M$, so that for an extremal hole ($j = 1$) a particle with the critical value $h = 2$ leaves the spin unchanged and particles with larger angular momentum are not absorbed at all (Cardoso et al. 2025). I mention this because it shows that the accretion of test particles cannot push a black hole past the Kerr bound, so that the extremal limit studied in Section 7.4 is the edge of the physical parameter space.

2.5. Circular photon orbits in the Kerr geometry

For $a \neq 0$ the single light ring of Schwarzschild splits into two equatorial circular photon orbits, one co-rotating (prograde) and one counter-rotating (retrograde) with respect to the hole. Their radii and angular velocities are (Bardeen et al. 1972)

$$\begin{aligned} r_{\rm ph}^{\pm} &= 2M\left\{1 + \cos\left[\tfrac{2}{3}\arccos\left(\mp\frac{a}{M}\right)\right]\right\}, \\ \Omega^{\pm} &= \frac{M^{1/2}}{(r_{\rm ph}^{\pm})^{3/2} \pm aM^{1/2}}, \end{aligned} \tag{2.15}$$

where the upper sign refers to the prograde orbit and the lower sign to the retrograde one. I write $\Omega_{\rm pro} = \Omega^+$ and $\Omega_{\rm ret} = \Omega^-$. At $a = 0$ both orbits are at $3M$, and as $a \to M$ the prograde orbit moves in towards the horizon, $r_{\rm ph}^+ \to M$ with $\Omega_{\rm pro} \to 1/2M$, while the retrograde orbit moves out to $r_{\rm ph}^- \to 4M$ with $\Omega_{\rm ret} \to 1/7M$. These two frequencies, and not the single value Ω_c, are the ones that the quasinormal frequencies follow at large ℓ for co-rotating and counter-rotating modes, as shown in Section 7.2. Figure 2.1 shows these quantities as functions of spin.

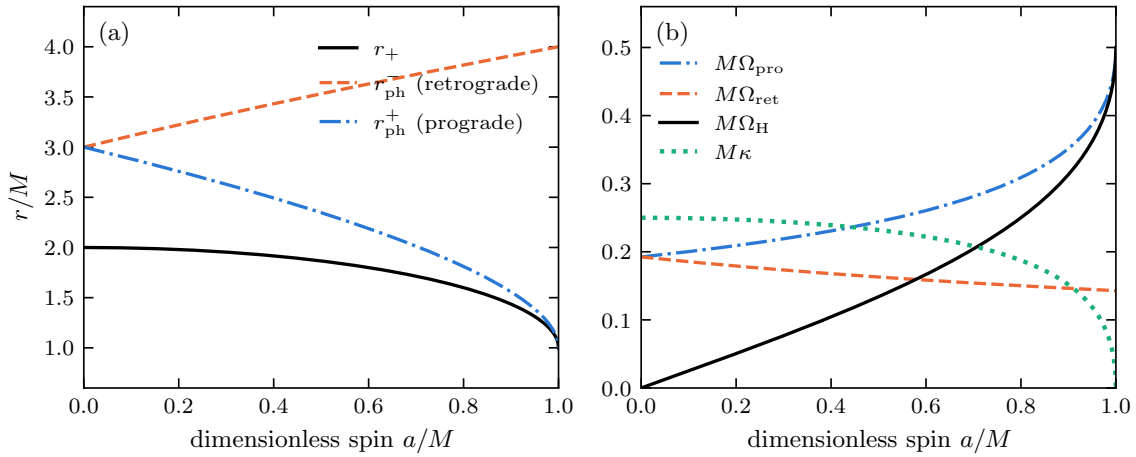


Figure 2.1.: Kerr geometry as a function of the dimensionless spin a/M ($M = 1$). (a) Radii of the outer horizon r_+ (solid) and of the prograde and retrograde equatorial photon orbits (dashed). (b) Angular velocities $\Omega_{\rm pro}$, $\Omega_{\rm ret}$ and $\Omega_{\rm H}$ of the photon orbits and of the horizon, and the surface gravity κ.

3. Linear perturbations of black holes

The geometry of Chapter 2 is a solution of the vacuum Einstein equations, and the question that defines this dissertation is how it responds to small disturbances. This chapter formulates that problem. It follows Part II of Cardoso et al. (2025) for the scalar, electromagnetic and gravitational master equations, scattering and superradiance, Chandrasekhar (1983) for the relation between the odd and even gravitational sectors, and Teukolsky (1973) for the rotating case.

3.1. The perturbative framework

Consider first a minimally coupled scalar field Φ of mass μ on a black-hole background, described by the action

$$S = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{R}{16\pi} - \frac{1}{2}g^{\mu\nu}\partial_\mu\Phi\,\partial_\nu\Phi - \frac{1}{2}\mu^2\Phi^2\right], \tag{3.1}$$

in units $G = c = 1$. Varying with respect to the field gives the Klein–Gordon equation $\Box\Phi = \mu^2\Phi$, and varying with respect to the metric gives the Einstein equations with a source that is quadratic in Φ. If the field is small, of order ϵ, the source is of order ϵ^2 and can be neglected at leading order, so that the metric remains the vacuum solution, Schwarzschild or Kerr, and the only equation left to solve is the Klein–Gordon equation on this fixed background. The same logic applies to an electromagnetic field. For gravitational perturbations the unknown is the metric itself, $g_{\mu\nu} = g^{(0)}_{\mu\nu} + h_{\mu\nu}$, and the vacuum equations are linearised in $h_{\mu\nu}$, giving $\delta R_{\mu\nu}[h] = 0$. These equations are invariant under the gauge transformations $h_{\mu\nu} \to h_{\mu\nu} + \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu$, which reflect the freedom to relabel the coordinates, and a gauge choice has to be made before they can be solved.

3.2. Master equations on the Schwarzschild background

The spherical symmetry of the background allows the angular dependence to be separated. A scalar field is expanded in spherical harmonics, $\Phi = \sum_{\ell m} \Psi_{\ell m}(t,r)\, Y_{\ell m}(\theta,\phi)/r$, and the Klein–Gordon equation for each (ℓ, m) becomes a wave equation in t and r only. In terms of the tortoise coordinate (2.4) it takes the form

$$\frac{\partial^2\Psi}{\partial r_*^2} - \frac{\partial^2\Psi}{\partial t^2} - V_s\Psi = 0, \qquad V_s = f\left[\frac{\ell(\ell+1)}{r^2} + \frac{2M(1-s^2)}{r^3}\right], \tag{3.2}$$

with $s = 0$ for a massless scalar field. The potential vanishes at both ends, because $f \to 0$ at the horizon and $\ell(\ell+1)/r^2 \to 0$ at infinity, and so in both limits the equation is a free wave equation and the solutions are plane waves $e^{\mp \mathrm{i}\omega r_*}$.

Vector and tensor fields require vector and tensor spherical harmonics, which are built from $Y_{\ell m}$ by taking its gradient, its curl and its projection along r. They separate into two classes with different behaviour under the parity transformation $(\theta,\phi) \to (\pi-\theta, \phi+\pi)$, called odd (or axial, magnetic-type) and even (or polar, electric-type). For the electromagnetic field the two classes reduce to the same equation (3.2) with $s = 1$, whose potential is $V_1 = f\,\ell(\ell+1)/r^2$. For gravitational perturbations, expanding $h_{\mu\nu}$ in tensor harmonics and working in the Regge–Wheeler gauge, in which the freedom in the angular part of the perturbation is used up to remove redundant functions, the ten components of the linearised Einstein equations reduce to exactly two independent master functions, one for each parity, which correspond to the two polarisations of a gravitational wave.

For odd parity the master function $\Psi = fh_1/r$ obeys equation (3.2) with $s = 2$, which is the Regge–Wheeler equation (Regge and Wheeler 1957),

$$V_{\mathrm{RW}} = f\left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3}\right]. \tag{3.3}$$

I derived this equation from the linearised Ricci tensor with a computer-algebra system, and the derivation is given in Appendix A. As a check of the machinery, a pure-gauge perturbation gave a vanishing $\delta R_{\mu\nu}$, and the potential that results coincides identically with the form (3.3) in the literature. For even parity the cor-

responding equation has the Zerilli potential (Zerilli 1970)

$$V_{\rm Z} = \frac{2f}{r^3}\,\frac{9M^3 + 9\lambda M^2 r + 3\lambda^2 M r^2 + \lambda^2(1+\lambda)r^3}{(\lambda r + 3M)^2}, \qquad \lambda = \frac{(\ell-1)(\ell+2)}{2}, \quad (3.4)$$

which is more complicated and has no obvious relation to equation (3.3). The angular number is restricted to $\ell \geq s$, since the lower multipoles of the gravitational and electromagnetic fields are not dynamical (Section 3.5).

3.3. Isospectrality of the odd and even sectors

It is a remarkable fact that two such different potentials lead to the same quasinormal spectrum. The reason is that they are superpartners in the sense of supersymmetric quantum mechanics (Chandrasekhar 1983). Let $\mu^2 = (\ell-1)(\ell+2)$ and define

$$F = \frac{f}{r(\mu^2 r + 6M)}, \qquad W = 6M\,F + \Omega_{\rm AS}, \qquad \Omega_{\rm AS} = \frac{(\ell-1)\ell(\ell+1)(\ell+2)}{12M}. \quad (3.5)$$

Then, with $'$ denoting $\mathrm{d}/\mathrm{d}r_*$,

$$V_{\rm RW} = W^2 - W' - \Omega_{\rm AS}^2, \qquad V_{\rm Z} = W^2 + W' - \Omega_{\rm AS}^2. \quad (3.6)$$

This is Chandrasekhar's relation, and I verified symbolically that equations (3.5) and (3.6) reproduce the potentials (3.3) and (3.4) for general ℓ.

The consequence for the spectrum follows from a short argument. Define the operator $\mathcal{A} = \mathrm{d}/\mathrm{d}r_* + W$ and its adjoint $\mathcal{A}^\dagger = -\mathrm{d}/\mathrm{d}r_* + W$. A direct calculation shows that $\mathcal{A}^\dagger\mathcal{A} = -\mathrm{d}^2/\mathrm{d}r_*^2 + W^2 - W'$ and $\mathcal{A}\mathcal{A}^\dagger = -\mathrm{d}^2/\mathrm{d}r_*^2 + W^2 + W'$, so that the Regge–Wheeler and Zerilli operators are $\mathcal{A}^\dagger\mathcal{A} - \Omega_{\rm AS}^2$ and $\mathcal{A}\mathcal{A}^\dagger - \Omega_{\rm AS}^2$. If Ψ^- is a solution of the Regge–Wheeler equation at frequency ω, then $\Psi^+ = \mathcal{A}\Psi^-$ is a solution of the Zerilli equation at the same frequency, and conversely $\Psi^- \propto \mathcal{A}^\dagger\Psi^+$, with $\mathcal{A}^\dagger\mathcal{A}\Psi^- = (\omega^2 + \Omega_{\rm AS}^2)\Psi^-$. Since $W \to \Omega_{\rm AS}$ at both ends, a solution that behaves as $e^{-\mathrm{i}\omega r_*}$ at the horizon is mapped to one that behaves as $(\Omega_{\rm AS} - \mathrm{i}\omega)\,e^{-\mathrm{i}\omega r_*}$, and a solution that behaves as $e^{+\mathrm{i}\omega r_*}$ at infinity to $(\Omega_{\rm AS} + \mathrm{i}\omega)\,e^{+\mathrm{i}\omega r_*}$. The quasinormal boundary conditions of Section 4.2 are therefore preserved by the transformation, with a non-zero constant, unless $\omega = \pm\mathrm{i}\Omega_{\rm AS}$. The two sectors then have identical spectra. The exceptional value is the algebraically special frequency of Chandrasekhar (1984),

$$M\omega_{\rm AS} = -\mathrm{i}\,\frac{(\ell-1)\ell(\ell+1)(\ell+2)}{12}, \quad (3.7)$$

which equals $-2\mathrm{i}$ for $\ell = 2$ and $-10\mathrm{i}$ for $\ell = 3$. At this frequency the transformation degenerates, and the two sectors are not guaranteed to agree. The behaviour of the spectrum near it is discussed in Section 4.5 and again in Chapter 6. Because of isospectrality, only the Regge–Wheeler equation needs to be solved in the numerical work.

3.4. The potential barrier and the light ring

For all multipoles $\ell \geq 2$ the potential V_s is a single barrier, which vanishes at both ends and has one maximum in between. Setting $\mathrm{d}V_s/\mathrm{d}r = 0$ and expanding in powers of $1/\ell(\ell+1)$ gives, with a symbolic computation, the position of the maximum,

$$r_{\mathrm{peak}} = 3M + \frac{(s^2-1)M}{3\,\ell(\ell+1)} + \frac{8(s^2-1)^2 M}{27\,[\ell(\ell+1)]^2} + \cdots , \tag{3.8}$$

while the height of the barrier tends to $V_{\mathrm{peak}} \to \ell(\ell+1)/27M^2$. For large ℓ the potential therefore converges to the null-geodesic potential of equation (2.7), with $L^2 \to \ell(\ell+1)$, and the maximum approaches the light ring from outside for $s = 2$. This is the first connection between waves and null geodesics. The barrier peaks near the light ring, and in the limit of high angular momentum wave propagation reduces to the motion of photons. The second connection, the identification of the quasinormal frequencies at large ℓ with the orbital frequency and Lyapunov exponent of the light ring, is developed in Section 4.5. The convergence is slow for low multipoles: for $\ell = 2$ the peak of V_{RW} lies at $3.281M$ and has only 68 per cent of the eikonal height (Table 6.1).

A second consequence is the reflection of waves. A wave with $M\omega \lesssim \ell/(3\sqrt{3})$ has less energy than the peak of the barrier and is mostly reflected, whereas a wave of higher frequency passes over the barrier and is mostly absorbed. This is the origin of the two limits of the absorption cross-section discussed in Section 3.6.

3.5. Static perturbations

In the static limit $\partial_t \Psi = 0$ the master equation becomes an ordinary differential equation,

$$f^2\Psi'' + ff'\Psi' - V_s\Psi = 0, \tag{3.9}$$

where now $' = \mathrm{d}/\mathrm{d}r$. Its two singular points, $r = 2M$ and $r = \infty$, are regular singular points, and the solutions behave there as $\Psi \sim A_+ + B_+ \ln(r-2M)$ and

$\Psi \sim A_\infty r^{-\ell} + B_\infty r^{\ell+1}$. A regular solution requires $B_+ = B_\infty = 0$, and there is no reason why this should be possible simultaneously. Multiplying the equation by Ψ^*/f and integrating over the exterior shows that it is not possible when $V_s > 0$. Regularity removes the boundary terms and what remains is

$$\int_{2M}^{\infty} f\,|\Psi'|^2\,\mathrm{d}r + \int_{2M}^{\infty} \frac{V_s}{f}\,|\Psi|^2\,\mathrm{d}r = 0, \tag{3.10}$$

so that $\Psi \equiv 0$ (Cardoso et al. 2025). There is no static scalar hair, and the same holds for the fields of higher spin with the exception of low multipoles. For the electromagnetic field with $\ell = 0$ the potential vanishes and the static solution is a shift of the charge of the hole. For the odd gravitational perturbation with $\ell = 1$ the regular solution is $\Psi = k/r$, which corresponds to adding angular momentum, and the lowest even-parity multipoles correspond to a change of the mass and a boost. These are changes to the parameters of the background, and not independent degrees of freedom.

A related property is that the black hole does not polarise. For a conducting sphere of radius R in a uniform electric field the potential is $U = -Er(1-R^3/r^3)\cos\theta$, and the second term is the induced dipole. The static scalar equation on Schwarzschild is solved by Legendre functions, $\Psi = C_1\, rP_\ell(r/M-1) + C_2\, rQ_\ell(r/M-1)$. The function Q_ℓ diverges logarithmically at the horizon, so that $C_2 = 0$, and the remaining solution is a polynomial in r with no term that decays as $r^{-\ell-1}$. The static tidal Love numbers, which would measure such a response, therefore vanish (Cardoso et al. 2025). I mention these results because they complete the picture of the rigidity of the geometry that underlies the no-hair argument. They are not used in the numerical work.

3.6. Scattering and absorption

The potential barrier determines how a black hole scatters a monochromatic wave. For real ω the solution of

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}r_*^2} + \left(\omega^2 - V_s\right)\psi = 0 \tag{3.11}$$

that is purely ingoing at the horizon, as it must be for a classical black hole, has the asymptotic form

$$\psi \to \begin{cases} e^{-\mathrm{i}\omega r_*}, & r_* \to -\infty, \\ A_{\mathrm{in}}\, e^{-\mathrm{i}\omega r_*} + A_{\mathrm{out}}\, e^{+\mathrm{i}\omega r_*}, & r_* \to +\infty. \end{cases} \tag{3.12}$$

Because the potential is real, the Wronskian of ψ and ψ^* is independent of r_*. Evaluating it at both ends gives

$$|A_{\rm in}|^2 - |A_{\rm out}|^2 = 1, \tag{3.13}$$

which states conservation of energy: the incident flux is the sum of the reflected flux and the flux that falls through the horizon. The reflection coefficient $|A_{\rm out}/A_{\rm in}|^2$ is therefore smaller than one at every frequency.

For a plane scalar wave of frequency ω the absorption cross-section is obtained by summing over partial waves,

$$\sigma_{\rm abs} = \sum_\ell \frac{\pi(2\ell+1)}{\omega^2}\left(1 - \frac{|A_{\rm out}|^2}{|A_{\rm in}|^2}\right). \tag{3.14}$$

Two limits are exactly known. At low frequency, $M\omega \ll 1$, only $\ell = 0$ contributes and the equation can be solved by matching a near-horizon solution $R \propto (r - 2M)^{-2{\rm i}M\omega}$ to a far-zone solution in spherical Bessel functions in the overlap region $2M \ll r \ll 1/\omega$. This gives $|A_{\rm out}/A_{\rm in}|^2 = 1 - 16(M\omega)^2$ and hence $\sigma_{\rm abs} = 16\pi M^2$, which is the area of the horizon. At high frequency many multipoles contribute and the cross-section tends to the geometrical value $27\pi M^2$ of equation (2.8). The black hole behaves as a sphere of radius $b_c = 3\sqrt{3}M$ for waves that are short compared with its size, and as its horizon for long waves.

3.7. Rotation and superradiance

In Kerr only the axisymmetry survives and the angular dependence is more complicated (Section 3.8), but the radial equation can still be brought to the form of a scattering problem,

$$\frac{{\rm d}^2\psi}{{\rm d}r_*^2} + \left[(\omega - m\varpi)^2 - V\right]\psi = 0, \qquad \varpi = \frac{a}{r^2+a^2}, \tag{3.15}$$

where m is the azimuthal number and $\varpi(r_+) = \Omega_{\rm H}$. The horizon boundary condition is that the wave is ingoing in the frame that co-rotates with the horizon, $\psi \sim A_T\, e^{-{\rm i}(\omega - m\Omega_{\rm H})r_*}$. Repeating the Wronskian argument that led to equation (3.13) now gives

$$|A_{\rm in}|^2 - |A_{\rm out}|^2 = \frac{\omega - m\Omega_{\rm H}}{\omega}\,|A_T|^2. \tag{3.16}$$

The right-hand side is negative if

$$\omega < m\Omega_{\rm H}, \tag{3.17}$$

and then $|A_{\rm out}| > |A_{\rm in}|$: the reflected wave has more energy than the incident one. This is superradiant scattering (Bardeen et al. 1972; Brito et al. 2015; Press and Teukolsky 1972), and the energy is taken from the rotation of the hole. It is the wave counterpart of the Penrose process of Section 2.4.1, and it requires the ergoregion.

It is not an accident of the wave equation. It follows from the area theorem alone. Suppose a wave of frequency ω and azimuthal number m is partly absorbed by a body that rotates with angular velocity Ω. The wave carries angular momentum $\mathrm{d}J = (m/\omega)\mathrm{d}E$, so that the change of the energy measured in the frame that corotates with the body is $\mathrm{d}E_0 = \mathrm{d}E - \Omega\,\mathrm{d}J = \mathrm{d}E\,(1 - m\Omega/\omega)$. By the second law the entropy $\mathrm{d}S = \mathrm{d}E_0/T$ cannot decrease, which requires $(\omega - m\Omega)Z_m \geq 0$ with Z_m the absorbed fraction. For $\omega < m\Omega$ this forces $Z_m < 0$, so that the body emits more than it absorbs (Cardoso et al. 2025). The condition (3.17) is relevant for quasinormal modes because the real parts of the near-extremal Kerr frequencies approach $m\Omega_{\rm H}$, the edge of the superradiant band (Section 7.4).

3.8. The Teukolsky equation

For a rotating black hole parity is no longer a good quantum number, and the separation of Section 3.2 into odd and even sectors is lost. Teukolsky (1973) showed, using the Newman–Penrose formalism, that the perturbations of Kerr can nevertheless be described by a single master equation for fields of spin weight $s = 0, \pm1, \pm2$ that separates in the coordinates. For gravitational perturbations the master variables are the Weyl scalars ψ_0 ($s = +2$) and ψ_4 ($s = -2$), suitably rescaled. With the time dependence $e^{-\mathrm{i}\omega t}e^{\mathrm{i}m\phi}$ the field separates as $\Psi = S(\theta)R(r)\,e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}$ with an angular equation

$$\begin{aligned}&\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S}{\mathrm{d}\theta}\right)\\ &\qquad+\left[a^2\omega^2\cos^2\theta-\frac{m^2}{\sin^2\theta}-2a\omega s\cos\theta-\frac{2ms\cos\theta}{\sin^2\theta}-s^2\cot^2\theta+s+A\right]S=0,\end{aligned} \tag{3.18}$$

whose regular solutions are the spin-weighted spheroidal harmonics, and a radial equation

$$\Delta^{-s}\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta^{s+1}\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\left[\frac{K^2-2\mathrm{i}s(r-M)K}{\Delta}+4\mathrm{i}s\omega r-\lambda\right]R=0, \tag{3.19}$$

with $K=(r^2+a^2)\omega-am$ and $\lambda=A+a^2\omega^2-2am\omega$. The angular eigenvalue A depends on the product $a\omega$, and it enters the radial equation through λ. For $a=0$ the spheroidal harmonics reduce to the spin-weighted spherical harmonics, $A=(\ell-s)(\ell+s+1)$, and for $s=-2$ equation (3.19) describes the same physics as the Regge–Wheeler and Zerilli equations.

The structural difference from the Schwarzschild case is therefore that the angular and radial problems are coupled through the unknown frequency. One cannot solve the angular problem once and then the radial problem for all frequencies: a quasinormal frequency has to satisfy both equations at once, and numerical methods have to iterate between them. This is the reason that the computation of Kerr modes is considerably harder than the Schwarzschild case, and is why Kokkotas and Schmidt (1999) remark that the first calculations were few in number. Rotation also complicates the formal analysis. Inside the ergoregion the stationary Killing vector is spacelike, and the equation obtained after a Fourier or Laplace transform in time is no longer elliptic, so that the construction of the Green function is correspondingly more involved (Kokkotas and Schmidt 1999). The boundary conditions are again ingoing at the horizon, with behaviour set by $\omega-m\Omega_{\mathrm{H}}$, and outgoing at infinity. Because m now enters the equations, the modes are labelled $\omega_{\ell mn}(a)$, and the $(2\ell+1)$-fold degeneracy of the Schwarzschild modes is lifted by rotation.

4. Quasinormal modes

The previous chapter set up the equations. This chapter defines the quasinormal modes as solutions of them, explains why they are complex and discrete, shows how they appear in the response of a black hole to a perturbation, and summarises the structure of the Schwarzschild and Kerr spectra that the computations of Chapters 6 and 7 are designed to test.

4.1. Why the frequencies are complex

A normal mode of a conservative system, such as a string fixed at both ends or a cavity filled with radiation, has a real frequency. The boundary conditions confine the energy, the spatial operator is self-adjoint on a Hilbert space, and a self-adjoint operator has a real spectrum and a complete set of eigenfunctions. A black hole is an open system. Energy leaves it in two ways, by falling through the horizon and by radiation to infinity, and neither end of the potential of Section 3.4 is a wall. There is no confinement, the operator is not self-adjoint with respect to the natural inner product, and nothing forces its eigenvalues to be real. The frequencies become complex, with the real part giving the rate of oscillation and the imaginary part the rate at which the energy leaks out (Kokkotas and Schmidt 1999). In the convention used here

$$\Psi(t, r_*) \propto e^{-\mathrm{i}\omega t} = e^{-\mathrm{i}\,\mathrm{Re}\omega\, t}\, e^{-|\mathrm{Im}\omega|\, t}. \tag{4.1}$$

This is also the reason for the name: the modes are *quasi*-normal because they decay.

4.2. Definition of the modes

To see how the modes arise from the initial-value problem, consider the master equation (3.2) with initial data $\Psi(0, r_*)$ and $\partial_t \Psi(0, r_*)$, and take the Laplace transform $\hat{\psi}(r_*, \omega) = \int_0^\infty \Psi\, e^{\mathrm{i}\omega t}\, \mathrm{d}t$, which converges for $\mathrm{Im}\omega > 0$. The time derivatives become algebraic and the equation becomes an ordinary differential equation with a source

made of the initial data,

$$\frac{\mathrm{d}^2\hat{\psi}}{\mathrm{d}r_*^2} + \big(\omega^2 - V\big)\,\hat{\psi} = \mathcal{I}(r_*,\omega), \qquad \mathcal{I} = \mathrm{i}\omega\,\Psi(0,r_*) - \partial_t\Psi(0,r_*). \tag{4.2}$$

Let ψ_H be the solution of the homogeneous equation that is ingoing at the horizon and ψ_∞ the solution that is outgoing at infinity,

$$\psi_H \to \begin{cases} e^{-\mathrm{i}\omega r_*}, & r_* \to -\infty, \\ A_{\mathrm{in}}\, e^{-\mathrm{i}\omega r_*} + A_{\mathrm{out}}\, e^{+\mathrm{i}\omega r_*}, & r_* \to +\infty, \end{cases} \qquad \psi_\infty \to e^{+\mathrm{i}\omega r_*}, \quad r_* \to +\infty. \tag{4.3}$$

The Wronskian of the two is independent of r_*, and evaluating it at large r_* gives $W = \psi_H\psi_\infty' - \psi_H'\psi_\infty = 2\mathrm{i}\omega A_{\mathrm{in}}(\omega)$. The solution of equation (4.2) by variation of parameters is

$$\hat{\psi}(r_*,\omega) = \int_{-\infty}^{\infty} G(r_*,r_*';\omega)\,\mathcal{I}(r_*',\omega)\,\mathrm{d}r_*', \qquad G(r_*,r_*';\omega) = \frac{\psi_H(r_{*<})\,\psi_\infty(r_{*>})}{2\mathrm{i}\omega A_{\mathrm{in}}(\omega)}, \tag{4.4}$$

where $r_{*<} = \min(r_*, r_*')$ and $r_{*>} = \max(r_*, r_*')$. This Green function represents a wave that falls into the hole on the left of the source and radiates away on its right. It has a singularity wherever the denominator vanishes,

$$A_{\mathrm{in}}(\omega) = 0. \tag{4.5}$$

At such a frequency the solution that is ingoing at the horizon contains no ingoing component at infinity, and the equation has a solution with

$$\Psi \sim \begin{cases} e^{-\mathrm{i}\omega r_*}, & r_* \to -\infty, \\ e^{+\mathrm{i}\omega r_*}, & r_* \to +\infty. \end{cases} \tag{4.6}$$

These are the quasinormal modes: solutions that are purely ingoing at the horizon and purely outgoing at infinity, with no radiation coming in from either side. The frequencies satisfying equation (4.5) are the poles of the Green function.

The spectrum is discrete, and the reason is the same as for the quantisation of bound states in quantum mechanics. A second-order equation has a two-dimensional space of solutions, and the condition at the horizon fixes the solution up to a constant. The second condition, that no ingoing wave is present at infinity, is then one complex condition on one complex unknown, and it can be met only at isolated values of ω.

Since V is real, if Ψ satisfies the equation with boundary conditions (4.6) at frequency ω, its complex conjugate satisfies the equation at frequency $-\omega^*$ with the same boundary conditions. The frequencies therefore come in pairs $(\omega, -\omega^*)$, which are mirror images in the imaginary axis of the complex plane. The two members describe the same real physical mode, and only the one with $\mathrm{Re}\omega > 0$ is quoted. A frequency with $\mathrm{Im}\omega < 0$ gives an eigenfunction that grows at both ends, since $|e^{\pm \mathrm{i}\omega r_*}| = e^{\mp \mathrm{Im}\omega\, r_*}$ grows as $|r_*| \to \infty$. The quasinormal modes are therefore not normalisable and not members of the usual Hilbert space.

The modes are labelled by the angular numbers and an overtone number,

$$\omega_{\ell m n}, \qquad n = 0, 1, 2, \ldots, \tag{4.7}$$

where n orders the modes of given (ℓ, m) by increasing damping, so that $n = 0$ is the fundamental mode, the one that decays most slowly. In Schwarzschild the frequencies do not depend on m, so that each (ℓ, n) is $(2\ell + 1)$-fold degenerate, which is the consequence of the spherical symmetry. In Kerr the degeneracy is lifted.

4.3. From the modes to the signal

The time-domain response is obtained from the inverse Laplace transform,

$$\Psi(t, r_*) = \frac{1}{2\pi} \int_{-\infty+\mathrm{i}c}^{+\infty+\mathrm{i}c} \hat{\psi}(r_*, \omega)\, e^{-\mathrm{i}\omega t}\, \mathrm{d}\omega, \tag{4.8}$$

along a line above all the singularities of $\hat{\psi}$. For $t > 0$ the contour can be closed in the lower half of the complex plane. The poles of the Green function at the quasinormal frequencies give a contribution $\sum_n B_n e^{-\mathrm{i}\omega_n t}$, a sum over the modes of exponentially damped sinusoids, which is the ringdown of equation (1.1). The residues B_n depend on the initial data through the source $\mathcal{I}$ and are called the excitation coefficients. This dissertation does not compute them. The remaining parts of the contour are a large arc, which accounts for the prompt response, and a branch cut along the negative imaginary axis. The branch cut arises because the potential does not decay exponentially at large radius, but only as a power of r_*, and the associated integral decays as a power law in time (Kokkotas and Schmidt 1999; Leaver 1985).

The power-law tail was first found by Price (1972). For generic, compactly supported initial data its amplitude at fixed radius decays as $t^{-(2\ell+3)}$. If instead the multipole is present as a static field before the perturbation, the decay is

$t^{-(2\ell+2)}$ (Kokkotas and Schmidt 1999). For initial data that are time-symmetric, $\partial_t\Psi(0) = 0$, there is a further effect. The solution for given data can be written $\Psi(t) = \int[\,\mathcal{G}(t)\,\partial_t\Psi(0) + \partial_t\mathcal{G}(t)\,\Psi(0)\,]\mathrm{d}r'_*$, where $\mathcal{G}$ is the time-domain Green function, and so the first term, which carries the leading $t^{-(2\ell+3)}$ behaviour, is absent. The tail is then one power of t steeper. I confirm both exponents numerically in Section 6.5.

A signal made from a perturbed black hole therefore has three stages: a prompt burst, a ringdown in which the quasinormal modes dominate, and a power-law tail that eventually overtakes the exponentially decaying modes. Since the tail and the prompt response are not made of modes, the set of quasinormal modes is not complete, and the modes describe the response at intermediate times only.

4.4. Frequency, damping time and quality factor

The real part of a quasinormal frequency is the oscillation frequency, $f = \mathrm{Re}\omega/2\pi$, and the amplitude decays with an e-folding time $\tau = 1/|\mathrm{Im}\omega|$. The quality factor of equation (1.2), $Q = \pi f\tau$, is the useful measure of how many oscillations can be seen. Converting to physical units uses equation (1.3). Because the damping increases rapidly with the overtone number (Section 4.5), the overtones are much shorter lived than the fundamental mode, which is why the ringdown signal of a real binary is dominated by the least-damped mode. In the merger of two black holes the fundamental mode has $\ell = m = 2$ and $n = 0$, and this is the mode followed through most of the Kerr calculations of Chapter 7.

4.5. Structure of the Schwarzschild spectrum

The spectrum of the Regge–Wheeler equation has the following main features, which are the subject of Chapter 6.

The overtone ladder. At fixed ℓ the frequencies form a sequence in the complex plane, starting at the fundamental mode and with increasing damping. The real part decreases at first with n, while the imaginary part grows roughly linearly. A numerical sample is given in Table 6.4.

Large ℓ: the eikonal limit. For large ℓ and small n the frequencies approach

$$M\omega_{\ell n} \simeq \frac{1}{3\sqrt{3}}\left[\left(\ell+\tfrac{1}{2}\right) - \mathrm{i}\left(n+\tfrac{1}{2}\right)\right], \tag{4.9}$$

that is, $\mathrm{Re}\omega = \Omega_c(\ell + \frac{1}{2})$ and $\mathrm{Im}\omega = -\lambda_{\mathrm{L}}(n + \frac{1}{2})$, where Ω_c and λ_{L} are the orbital frequency and the Lyapunov exponent of the light ring from Section 2.3 (Cardoso et al. 2009; Ferrari and Mashhoon 1984; Goebel 1972). The oscillation frequency of the mode is the rate at which a photon goes around the black hole, and the damping is the rate at which nearby photon orbits diverge from the unstable one. A derivation based on the WKB formula is given in Section 5.4. The first correction in $1/\ell$ is (Dolan and Ottewill 2009)

$$M\omega_{\ell n} = \frac{1}{3\sqrt{3}}\left[L - \mathrm{i}N + \frac{\beta/3 - \frac{5}{36}N^2 - \frac{115}{432}}{L} + O(L^{-2})\right], \qquad L = \ell + \tfrac{1}{2},\ N = n + \tfrac{1}{2}, \tag{4.10}$$

where $\beta = 1 - s^2$ is -3 for gravitational perturbations.

Highly damped modes. For very large n the real part does not tend to zero. It tends to a constant that does not depend on ℓ,

$$\lim_{n\to\infty} \mathrm{Re}(M\omega_n) = \frac{\ln 3}{8\pi} = 0.043712\ldots, \tag{4.11}$$

while the spacing of the imaginary parts tends to $1/4$, so that $\mathrm{Im}(M\omega_n) \simeq -(n + \frac{1}{2})/4$. The constant was found numerically by Nollert (1993), recognised as $\ln 3/8\pi$ by Hod (1998), and proved by Motl and Neitzke (2003) through a monodromy argument in which the radial equation is continued around the horizon in the complex r plane. In terms of the Hawking temperature $T_{\mathrm{H}} = 1/8\pi M$, the real part is $T_{\mathrm{H}} \ln 3$ and the spacing of the imaginary parts is $2\pi T_{\mathrm{H}}$. That a transcendental number such as $\ln 3$ emerges from a classical differential equation in which no 3 appears is striking, and it led Hod (1998) to propose a connection with the quantisation of the horizon area. The numerical work in this dissertation does not depend on any such interpretation. The leading correction to the real part is $\gamma_1(2n+1)^{-1/2}$, with a coefficient γ_1 that depends on ℓ, and Nollert (1993) gives $\gamma_1 = 0.343$, 0.7545 and 2.81 for $\ell = 2$, 3 and 6. The highly damped modes are controlled by the structure of the potential near the horizon and not by its peak. For the same reason the WKB method, which expands about the peak, cannot describe them.

The algebraically special frequency. The frequency $M\omega_{\mathrm{AS}} = -\mathrm{i}(\ell-1)\ell(\ell+1)(\ell+2)/12$ of equation (3.7) is the one at which the isospectrality argument of Section 3.3 fails. It is also where the Teukolsky–Starobinsky constant, which relates solutions of opposite spin weight, vanishes (Chandrasekhar 1984; Maassen van den Brink 2000). Kokkotas and Schmidt (1999) describe the status of this frequency as a quasinormal mode as unsettled, and mention the suggestion that the poten-

tial becomes transparent there. The practical consequence is that the numerical determination of the modes close to this frequency is delicate. I return to this in Section 6.4.

4.6. Structure expected in the Kerr spectrum

With rotation the modes are labelled by m as well, and their frequencies depend on the spin a/M. Three patterns are expected, and each is tested in Chapter 7.

First, the lifting of the degeneracy is similar to the splitting of atomic levels by a magnetic field, with the spin axis playing the role of the field. At small spin the shift of the real part is linear in ma, while at large spin co-rotating modes ($m > 0$) and counter-rotating modes ($m < 0$) behave very differently, as a consequence of the coupling $\omega \to \omega - m\Omega_{\mathrm{H}}$ in the radial equation of Section 3.7. Second, at large ℓ the real part of the frequency should follow the orbital frequency of the circular photon orbit, the prograde one for $m = +\ell$ and the retrograde one for $m = -\ell$, by the generalisation of equation (4.9) using the orbits of Section 2.5. Third, as $a \to M$ the surface gravity κ of equation (2.12) vanishes, and some of the modes lose their damping. Detweiler (1980) found that the damping of some modes becomes very small in this limit, and Hod (2008) derived their behaviour,

$$\omega \simeq m\Omega_{\mathrm{H}} - \mathrm{i}\,\kappa\left(n + \tfrac{1}{2}\right), \tag{4.12}$$

so that the real part approaches the edge of the superradiant band (3.17) and the damping vanishes with $\kappa = 2\pi T_{\mathrm{H}}$. Yang et al. (2013) showed that equation (4.12) describes one family of modes only, called zero-damping modes, and that a second family, the damped modes, retains a finite damping in the extremal limit.

4.7. Stability

A mode with $\mathrm{Im}\omega > 0$ would grow exponentially, and the existence of such a mode would mean that the black hole is unstable. Regge and Wheeler were in fact motivated by the stability question when they derived their equation, and the stability of Schwarzschild against linear gravitational perturbations is now established mathematically (Dafermos et al. 2019). For Kerr, the absence of growing modes (mode stability) was proved by Whiting (1989), with quantitative bounds by Shlapentokh-Rothman (2015), and linear stability of slowly rotating Kerr black holes was proved by Häfner et al. (2021). When Kokkotas and Schmidt (1999) wrote their review,

the stability of Kerr was still an open question. Every frequency computed in this dissertation has $\mathrm{Im}\omega < 0$, as it should.

5. Numerical methods

Closed-form quasinormal frequencies are known only in a few approximate limits. The spectra studied in this dissertation were therefore computed numerically, with four tools: Leaver's continued-fraction method for Schwarzschild, the WKB approximation and its eikonal limit as analytic cross-checks, the QNM package for Kerr, and a time-domain evolution of the Regge–Wheeler equation. This chapter describes each of them and the tests that establish their accuracy. The Schwarzschild solver (Sections 5.1–5.3) and the time-domain code (Section 5.6) were written for this project, and the symbolic derivations of the recurrence relation and of its asymptotic tail are my own.

5.1. Leaver's continued fraction for Schwarzschild

Units $2M = 1$ are used throughout this section, so that the horizon is at $r = 1$. The Regge–Wheeler equation (3.2) written in r is

$$f\frac{\mathrm{d}}{\mathrm{d}r}\left(f\frac{\mathrm{d}\Psi}{\mathrm{d}r}\right)+\left(\omega^2 - V\right)\Psi = 0, \qquad V = f\left[\frac{\ell(\ell+1)}{r^2}+\frac{1-s^2}{r^3}\right], \quad f = 1-\frac{1}{r}. \tag{5.1}$$

The method of Leaver (1985) starts from a solution that has the correct behaviour at both ends built in. Near the horizon the tortoise coordinate is $r_* = r + \ln(r-1)$, so that the ingoing condition $e^{-\mathrm{i}\omega r_*}$ corresponds to $(r-1)^{-\mathrm{i}\omega}$ times a function regular at $r = 1$, and at large r the outgoing condition $e^{+\mathrm{i}\omega r_*}$ corresponds to $r^{\mathrm{i}\omega}e^{\mathrm{i}\omega r}$. Both requirements are met by the ansatz

$$\Psi(r) = (r-1)^{-\mathrm{i}\omega}\, r^{2\mathrm{i}\omega}\, e^{\mathrm{i}\omega(r-1)} \sum_{n=0}^{\infty} a_n \left(\frac{r-1}{r}\right)^n, \tag{5.2}$$

where the prefactor behaves as $(r-1)^{-\mathrm{i}\omega}$ at the horizon and as $r^{\mathrm{i}\omega}e^{\mathrm{i}\omega r}$ at infinity. The compactified variable $x = (r-1)/r$ maps the whole exterior to the interval $0 \le x < 1$. Substituting the ansatz into the equation, I obtained with a computer-algebra system (Appendix B) that the series coefficients satisfy a three-term recurrence relation.

With $\rho = -\mathrm{i}\omega$ and $\varepsilon = s^2 - 1$,

$$\alpha_n a_{n+1} + \beta_n a_n + \gamma_n a_{n-1} = 0, \qquad n \geq 1, \qquad \alpha_0 a_1 + \beta_0 a_0 = 0, \tag{5.3}$$

with

$$\begin{aligned} \alpha_n &= n^2 + (2\rho + 2)n + 2\rho + 1, \\ \beta_n &= -\left[2n^2 + (8\rho + 2)n + 8\rho^2 + 4\rho + \ell(\ell+1) - \varepsilon\right], \\ \gamma_n &= n^2 + 4\rho n + 4\rho^2 - s^2. \end{aligned} \tag{5.4}$$

Although the series is constructed to satisfy the condition at the horizon term by term, it satisfies the condition at infinity only if it converges at $x = 1$, where $r \to \infty$. A three-term recurrence has two independent solutions. For large n their ratios behave as $a_{n+1}/a_n \to 1 \pm \sqrt{2\rho}/\sqrt{n}$, so that one solution decays as $\exp(-2\sqrt{2\rho n})$, is the *minimal* solution, and gives a convergent series at $x = 1$, while the other is dominant and gives a divergent one. A divergent series corresponds to an ingoing wave at infinity. The quasinormal condition is therefore that the sequence a_n generated by the starting relation $\alpha_0 a_1 + \beta_0 a_0 = 0$ is the minimal solution. By Pincherle's theorem this holds if and only if the ratio a_{n+1}/a_n is given by the continued fraction obtained by solving the recurrence downwards, and the starting relation then becomes an equation for the frequency,

$$\beta_0 - \frac{\alpha_0\gamma_1}{\beta_1 -}\,\frac{\alpha_1\gamma_2}{\beta_2 -}\,\frac{\alpha_2\gamma_3}{\beta_3 -}\cdots = 0. \tag{5.5}$$

Its roots are the quasinormal frequencies $\omega_{\ell n}$ for the given ℓ and s.

For the nth overtone it is better to use an inverted form of this equation. The equation can be rewritten as the recurrence relation at index n,

$$\beta_n + \gamma_n \frac{a_{n-1}}{a_n} + \alpha_n \frac{a_{n+1}}{a_n} = 0, \tag{5.6}$$

in which a_{n-1}/a_n is obtained by forward recursion from the starting relation, and a_{n+1}/a_n by backward recursion from the minimal solution. This is Leaver's nth inversion. The inversion improves the numerical stability of the root search for the higher overtones, because it makes the nth root the one that the equation determines most strongly (Leaver 1985; Nollert 1993).

5.1.1. The remainder of the continued fraction

In practice the continued fraction is truncated at a large depth N, and the truncation error is large unless the tail is estimated. If the ratio is simply set to zero at the cutoff, a negligible-looking error of the tail translates into an error in the frequency that decreases slowly with N for modes with large $|\omega|$ (Section 5.3). Nollert (1993) proposed to replace the vanishing remainder by the asymptotic series

$$\frac{a_{k+1}}{a_k} = 1 + \frac{C_1}{\sqrt{k}} + \frac{C_2}{k} + \frac{C_3}{k^{3/2}} + \frac{C_4}{k^2} + O\left(k^{-5/2}\right), \tag{5.7}$$

which follows from the recurrence for large k. I derived the coefficients symbolically by substituting this series into the recurrence and equating powers of $k^{-1/2}$. The result, given in Appendix C, is $C_1 = \pm\sqrt{2\rho}$, $C_2 = 2\rho - \frac{3}{4}$ and expressions for C_3 and C_4 that extend the first two coefficients given by Nollert (1993). The sign of C_1 is chosen so that the series is the minimal solution, that is the ratio with the smaller modulus. At $N = 10^3$ and for the fundamental $\ell = 2$ mode the series differs from the ratio obtained by iterating the recurrence downwards from a very large index by 9×10^{-8}, which is consistent with the neglected $k^{-5/2}$ term.

5.2. Implementation and acceptance criteria

The roots of equation (5.6) were found with a secant iteration to a relative tolerance of 10^{-13} in double precision. The continued-fraction depth was $N = 3000 + 60n$, and every mode was computed again at a depth of $2N$. A mode is accepted only if

1. the iteration converged at both depths;
2. the two results differ by less than 10^{-8};
3. the imaginary part differs from that of the preceding overtone by an amount between 0.15 and 0.6, which guards against skipped or repeated modes, since the spacing of the imaginary parts is $1/4$ asymptotically.

The starting values for $n \leq 20$ were taken from the tabulation of Schwarzschild frequencies distributed with the QNM package (Stein 2019), and those for $n > 20$ by linear extrapolation from the two preceding overtones. The program refined each tabulated value, so that the tabulation serves as a seed and as an independent check, and the accepted frequencies do not depend on it. I computed the fundamental modes for $\ell = 2, 3, 4, 5, 10, 20$ and 100, and the overtones up to $n = 130$ for $\ell = 2$,

Table 5.1.: Convergence of the Schwarzschild $\ell = 2$ overtones with the truncation depth N of the continued fraction. Entries are $|\omega_N - \omega_{\text{ref}}|$ (in units of M^{-1}), where ω_{ref} is the value obtained at $N = 16\,000$ with the remainder. "Plain" means that the ratio a_{N+1}/a_N is set to zero, and "tail" that it is replaced by the series (5.7).

	$n = 5$		$n = 20$		$n = 60$	
N	plain	tail	plain	tail	plain	tail
25	1.3×10^{-2}	4.3×10^{-5}	5.5×10^{-2}	1.7×10^{-2}	–	–
50	4.1×10^{-3}	2.0×10^{-6}	1.8×10^{-1}	8.1×10^{-4}	–	–
100	6.2×10^{-4}	6.8×10^{-8}	2.3×10^{-2}	8.6×10^{-5}	6.5×10^{-2}	3.2×10^{-3}
200	4.6×10^{-5}	1.2×10^{-9}	1.5×10^{-2}	9.4×10^{-6}	2.9×10^{-2}	3.1×10^{-4}
400	1.2×10^{-6}	7.4×10^{-12}	5.7×10^{-3}	8.6×10^{-7}	1.8×10^{-2}	3.9×10^{-5}
800	7.0×10^{-9}	1.1×10^{-14}	1.5×10^{-3}	5.6×10^{-8}	1.5×10^{-2}	4.6×10^{-6}
1600	5.0×10^{-12}	$< 10^{-16}$	2.4×10^{-4}	2.2×10^{-9}	4.3×10^{-3}	4.5×10^{-7}
3200	2.2×10^{-16}	$< 10^{-16}$	1.9×10^{-5}	4.2×10^{-11}	1.5×10^{-3}	3.1×10^{-8}

$n = 90$ for $\ell = 3$ and $n = 60$ for $\ell = 4$, apart from the modes near the algebraically special frequency that did not pass the criteria (Section 6.4). The results are on the branch $\operatorname{Re}\omega > 0$, and the partner is $-\omega^*$.

5.3. Validation of the solver

Three tests were applied. The first compares the frequencies with the tabulation of the QNM package for $n \leq 20$. The largest differences are 7×10^{-7} for $\ell = 2$ (at $n = 20$), 1×10^{-10} for $\ell = 3$ and 2×10^{-10} for $\ell = 4$, which are within the uncertainties quoted with the tabulation, of 2×10^{-6} at $n = 20$ for $\ell = 2$ and 10^{-10} for the other two.

The second test is the convergence with depth. Table 5.1 lists, for $\ell = 2$, the difference between the frequency computed at depth N and the reference value obtained at $N = 16\,000$, for plain truncation of the continued fraction and for truncation with the remainder (5.7). For the fundamental mode both agree with the reference to rounding error from $N \simeq 100$. For the overtones the gain from the remainder grows with the depth. At $N = 3200$ it reduces the error by a factor of 4.5×10^5 for $n = 20$ and 4.8×10^4 for $n = 60$: plain truncation still has an error of 1.5×10^{-3} for $n = 60$, whereas the series gives 3×10^{-8}. The depth of $3000 + 60n$ used for the production runs is therefore enough for the acceptance criterion of 10^{-8} given above, which compares results at N and $2N$.

The third test is the potential itself, since the solver is only as good as the equation that it solves. The Regge–Wheeler potential used here was derived from the field

equations (Appendix A), and it coincides identically with the published form (3.3). In addition, the time-domain evolution of Section 5.6 uses the same potential and a completely different algorithm. Its agreement with the continued-fraction frequencies (Section 6.5) tests both.

5.4. The WKB approximation and the eikonal limit

The quasinormal-mode problem resembles a bound-state problem for the inverted barrier. Near its maximum at r_{*0} the potential is approximately $V \simeq V_0 + \frac{1}{2}V_0''(r_* - r_{*0})^2$ with $V_0'' < 0$, and the equation $\psi'' + (\omega^2 - V)\psi = 0$ becomes a parabolic-cylinder equation. Requiring that the solution is outgoing on both sides of the barrier gives the first-order WKB condition of Schutz and Will (1985),

$$\omega^2 = V_0 - \mathrm{i}\left(n + \tfrac{1}{2}\right)\sqrt{-2V_0''}, \tag{5.8}$$

where V_0'' is the second derivative with respect to r_*. Higher orders are available (Iyer and Will 1987), but equation (5.8) is sufficient for the comparison here. I evaluated it numerically for the Regge–Wheeler potential, with the maximum and the derivative found by high-precision arithmetic.

The eikonal limit follows from equation (5.8) with the light-ring potential of Section 2.3. For large ℓ the potential tends to $V = fL/r^2$ with $L = \ell(\ell+1) \to (\ell+\frac{1}{2})^2$. At $r = 3M$ it is $V_0 = L/27M^2$, and $V_{rr}'' = -2L/81M^4$, so that with $f(3M) = \frac{1}{3}$ the second derivative with respect to r_* is $V_0'' = f^2 V_{rr}'' = -2L/729M^4$. Equation (5.8) gives

$$\omega^2 = \frac{1}{27M^2}\left[L - 2\mathrm{i}\left(n + \tfrac{1}{2}\right)\sqrt{L}\right] \simeq \frac{1}{27M^2}\left[\sqrt{L} - \mathrm{i}\left(n + \tfrac{1}{2}\right)\right]^2, \tag{5.9}$$

up to a term of order unity that is negligible relative to L, which is the eikonal formula (4.9). The expansion of Dolan and Ottewill (2009), equation (4.10), extends this to order $1/\ell$, and I compare it with the exact frequencies in Chapter 6.

5.5. Kerr: the *qnm* package and spin sequences

For Kerr black holes the radial Teukolsky equation (3.19) can also be solved with Leaver's method. The substitution of a series analogous to equation (5.2) gives a three-term recurrence of the form (5.3), whose coefficients depend on ω, a, m, s and the angular separation constant A. The condition that the minimal solution exists

is again a continued-fraction equation, but now it has to be solved together with the angular equation (3.18), because A is an eigenvalue that depends on $a\omega$. The QNM package of Stein (2019) solves the two problems together. It obtains A and the angular eigenfunction from a spectral decomposition of the spheroidal harmonic in spin-weighted spherical harmonics, which turns the angular equation into a matrix eigenvalue problem (Cook and Zalutskiy 2014), and searches for the root of the radial continued fraction with the nth inversion, recomputing A at each iteration. The package follows a mode along a sequence of spins by taking the converged solution at one spin, extrapolating it to the next, and using the extrapolation as the starting guess. I used it to compute the gravitational ($s = -2$) frequencies, and I designed and analysed the scans described below.

The calculations are the following sequences, all starting from the Schwarzschild value at $a = 0$ and following a given (ℓ, m, n) to higher spin:

1. the co-rotating fundamental modes and the first four overtones, $m = \ell$, $n = 0$–4, for $\ell = 2$–10, up to $a = 0.998$ in steps of $\Delta a = 0.02$;

2. the fundamental modes, $m = \ell$, for $\ell = 2$–10, and the overtones $n = 1$–4 for $\ell = 2$, up to $a = 0.99999$, with an adaptive step that is reduced from 0.01 down to 10^{-7} close to extremality;

3. the counter-rotating fundamental modes, $m = -\ell$, for $\ell = 2$–10, up to $a = 0.998$;

4. the overtones $n = 0$–7 of $\ell = m = 2$ with a fine step of 5×10^{-4} up to $a = 0.998$;

5. the multiplet $m = -2, \ldots, 2$ of the fundamental $\ell = 2$ mode up to $a = 0.998$ with $\Delta a = 0.005$.

The starting values of the sequences provide a test of the package against my own solver: for the fifteen cases $\ell = 2$–4, $n = 0$–4, the $a = 0$ frequencies from the package agree with those of Section 5.2 to 4×10^{-11} or better, the largest difference being for $\ell = 2$, $n = 4$. The overtone label n is assigned at $a = 0$ and carried along the sequence by continuity. That label is exactly what can fail when two branches approach each other, and the check of this is part of the analysis in Section 7.5. To test the tracking I repeated the sequences of $n = 4$, 5 and 6 for $\ell = m = 2$ with the step reduced by a factor of four and, separately, with higher radial and angular resolution in the solver. The three versions agree to 10^{-4} after interpolation to common spins, and to 10^{-10} where the spins coincide. A near-extremal sequence for $\ell = 2$, $n = 2$ was found to mistrack, duplicating the $n = 3$ branch, and is not

used. The overtone ladder is restricted to $n \leq 7$ because the Schwarzschild starting value of the $\ell = 2$, $n = 8$ mode is unreliable (Section 6.4).

5.6. Time-domain evolution

The frequency-domain methods above assume the boundary conditions of Section 4.2. As an independent test, and to see the three stages of the response of Section 4.3 directly, I integrated the Regge–Wheeler equation in time. With $M = 1$,

$$\frac{\partial^2 \Psi}{\partial t^2} = \frac{\partial^2 \Psi}{\partial r_*^2} - V_{\mathrm{RW}}(r)\, \Psi, \tag{5.10}$$

on a uniform grid in r_* with spacing Δr_*, using the second-order leapfrog scheme

$$\Psi_j^{k+1} = 2\Psi_j^k - \Psi_j^{k-1} + \left(\frac{\Delta t}{\Delta r_*}\right)^2 \left(\Psi_{j+1}^k - 2\Psi_j^k + \Psi_{j-1}^k\right) - \Delta t^2\, V_j \Psi_j^k, \tag{5.11}$$

with $\Delta t = \Delta r_*/2$, which satisfies the Courant condition. The potential on the grid requires $r(r_*)$, the inverse of equation (2.4), which is computed with the Lambert W function. The initial data were a Gaussian $\Psi(0, r_*) = \exp[-(r_* - 10)^2/18]$ centred at $r_* = 10M$ with width $3M$, either with $\partial_t \Psi(0) = 0$ (time-symmetric data) or with $\partial_t \Psi(0) = \Psi(0)$ (generic data), and the signal was recorded at fixed $r_* = 20M$ and $40M$. The domain was $-700M \leq r_* \leq 1700M$ with the field set to zero at both ends, and the run was stopped at $t = 1400M$. By then no signal from either boundary can have reached the observers: a pulse travelling at unit speed from the initial position to the left boundary and back arrives at $r_* = 20M$ at $t \simeq 1430M$, and the potential is negligible near both ends, so there is no backscatter.

Two sets of runs were made. For the study of the ringdown I used double precision with $\Delta r_* = 0.1M$ and $0.05M$ and a shorter domain, which gives the signal to a relative precision of about 10^{-15} of the initial amplitude. The power-law tail, however, has an amplitude that falls below 10^{-14} of the initial pulse after $t \simeq 600M$, which is at the level of rounding error of double precision. For the tail I therefore used 80-bit extended-precision arithmetic with $\Delta r_* = 0.2M$ and $0.1M$. The results at the two resolutions agree, as shown in Section 6.5.

5.7. Software and reproducibility

All of the computations were done in PYTHON with NUMPY, SCIPY, MPMATH, SYMPY and MATPLOTLIB. The symbolic work includes the linearised Ricci tensor for the odd-parity perturbation (Appendix A), the recurrence relation (Appendix B), the asymptotic coefficients of Appendix C, the superpartner relation of Section 3.3, and the expansion (3.8) of the position of the potential maximum. The scripts are listed in Appendix D. All figures and tables are generated from stored output of the programs, and no number is entered by hand.

6. Results for Schwarzschild black holes

This chapter presents the gravitational quasinormal spectrum of the Schwarzschild black hole computed with the methods of Chapter 5. It begins with the potential barrier, then compares the exact fundamental modes with the analytic approximations, follows the overtones to high order, discusses the region near the algebraically special frequency, and finishes with the time-domain evolution, which gives an independent check on the frequencies and shows the late-time tail. All frequencies are in units of M^{-1} and are those with $\mathrm{Re}\,\omega > 0$.

6.1. The Regge–Wheeler barrier

Figure 6.1 shows V_{RW} for $\ell = 2$–5 as a function of the tortoise coordinate. In each case the potential is a single barrier that vanishes at both ends, as discussed in Section 3.4, with its maximum near the light ring and a height that grows as $\ell(\ell+1)$. Table 6.1 gives the position and the height of the maximum for several multipoles. The height relative to the eikonal value $\ell(\ell+1)/27M^2$ rises from 0.68 at $\ell = 2$ to 0.98 at $\ell = 10$ and 0.9998 at $\ell = 100$, and the position approaches $3M$ from above. The series (3.8), truncated after its second correction, reproduces the position of the maximum to $0.04M$ at $\ell = 2$ and to better than $10^{-3}M$ for $\ell \geq 4$, which is a check on the symbolic expansion. For $\ell = 2$ the peak lies at $r = 3.281M$, well away from the light ring, so the geometric-optics picture should not be expected to be accurate at the lowest multipole.

6.2. Fundamental modes: exact values, WKB and the eikonal expansion

Table 6.2 lists the fundamental ($n = 0$) frequencies from the continued fraction, together with the first-order WKB estimate of equation (5.8) and the eikonal expansion

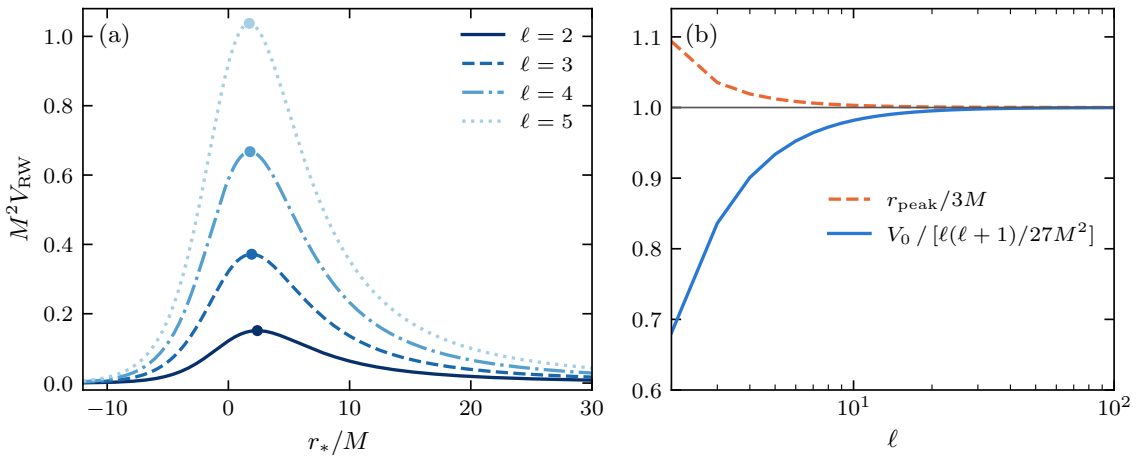


Figure 6.1.: The Regge–Wheeler potential for $M = 1$. (a) V_{RW} against the tortoise coordinate for $\ell = 2$–5. The dots mark the maxima. (b) Position and height of the maximum relative to their eikonal values, $3M$ and $\ell(\ell+1)/27M^2$, as a function of ℓ.

Table 6.1.: Maximum of the Regge–Wheeler potential ($M = 1$): the position r_{peak} from a numerical root of $\mathrm{d}V/\mathrm{d}r = 0$, the position from the series (3.8) truncated after the second correction, the height V_{peak}, and its ratio to the eikonal value $\ell(\ell+1)/27$.

ℓ	r_{peak}/M	r_{peak}/M (series)	$V_{\mathrm{peak}}M^2$	$V_{\mathrm{peak}}/\left[\ell(\ell+1)/27M^2\right]$
2	3.28078	3.24074	0.151287	0.6808
3	3.10611	3.10185	0.371625	0.8362
4	3.05753	3.05667	0.667360	0.9009
5	3.03654	3.03630	1.037481	0.9337
10	3.00932	3.00931	4.000115	0.9818
20	3.00240	3.00240	15.481511	0.9952
100	3.00010	3.00010	374.000001	0.9998

of equation (4.10). For $\ell = 2$ the exact frequency is $M\omega = 0.373672 - 0.088962\,\mathrm{i}$, which agrees with the standard tabulated value. The WKB real part overestimates the exact one by 6.7 per cent at $\ell = 2$, by 1.6 per cent at $\ell = 4$ and by 0.28 per cent at $\ell = 10$, and the imaginary part is better than 0.8 per cent already at $\ell = 2$. This is the typical accuracy of the lowest WKB order at low ℓ. It improves with ℓ because the barrier becomes narrower relative to its height, so that the expansion about the maximum is more accurate.

The leading-order eikonal result of equation (4.9) gives $\mathrm{Re}(M\omega) = (\ell+\frac{1}{2})/3\sqrt{3} = 0.481$ for $\ell = 2$, which is 29 per cent above the exact value. The expansion of Dolan and Ottewill (2009) with the $1/L$ correction is considerably better. Its real part is within 2.0 per cent of the exact value at $\ell = 2$, 0.43 per cent at $\ell = 3$, 0.15 per cent at $\ell = 4$ and 5×10^{-3} per cent at $\ell = 10$. The imaginary part is $-N/(3\sqrt{3})$ at this order, since the first correction to it is of order L^{-2}, and its errors are 8 per cent at

Table 6.2.: Schwarzschild fundamental mode ($n = 0$): exact frequencies from the continued fraction, the first-order WKB estimate of equation (5.8), and the relative errors (per cent) of the real part for the WKB estimate and for the eikonal expansion with the $1/L$ correction, equation (4.10). Frequencies are in units of M^{-1}.

ℓ	$\mathrm{Re}(M\omega)$	$-\mathrm{Im}(M\omega)$	$\mathrm{Re}(M\omega)_{\mathrm{WKB}}$	$-\mathrm{Im}(M\omega)_{\mathrm{WKB}}$	$\Delta\mathrm{Re}_{\mathrm{WKB}}$	$\Delta\mathrm{Re}_{\mathrm{eik}}$
2	0.373672	0.088962	0.398850	0.088285	+6.74	+1.956
3	0.599443	0.092703	0.616561	0.092318	+2.86	+0.434
4	0.809178	0.094164	0.822303	0.093929	+1.62	+0.150
5	1.012295	0.094871	1.022962	0.094714	+1.05	+0.065
10	1.996788	0.095864	2.002323	0.095821	+0.28	+0.005
20	3.933001	0.096131	3.935829	0.096120	+0.07	< 0.0005
100	19.338743	0.096221	19.339319	0.096221	< 0.005	< 0.0005

$\ell = 2$, falling to 0.4 per cent at $\ell = 10$. The correction is what makes the eikonal series useful at low ℓ: the real part, which is set by the light ring of Chapter 2, is reproduced to better than 0.5 per cent already at $\ell = 3$.

The two approximations behave very differently for the overtones. Table 6.3 compares them with the exact frequencies for $n = 0, 1, 2$. The error of the first-order WKB formula increases rapidly with n: for $\ell = 2$ the real part is wrong by 31 per cent at $n = 1$ and by 72 per cent at $n = 2$, and the imaginary part by 15 and 29 per cent. Even at $\ell = 4$ the error in the real part reaches 19 per cent at $n = 2$. The reason is that the formula is a local expansion about the top of the barrier, whereas a mode with a larger damping samples the potential further from the peak. In contrast, the real part of the eikonal expansion with the $1/L$ correction stays within 5.2 per cent for $\ell = 2$ and within 0.9 per cent for $\ell = 3$ and 4 up to $n = 2$. The eikonal expansion in $1/L$ is therefore the more reliable analytic estimate for the lowest overtones.

6.3. Overtones and the highly damped modes

Figure 6.2 shows the overtones of $\ell = 2$, 3 and 4, and Table 6.4 gives selected values. The modes with $\mathrm{Re}\,\omega > 0$ lie on a curve in the complex plane (panel a) that descends from the fundamental mode, and the mirror modes at $-\omega^*$ are omitted. The spacing of the imaginary parts is not uniform for the lowest overtones: for $\ell = 2$ it grows from 0.185 between $n = 0$ and 1 to 0.227 between $n = 2$ and 3. Over $n = 100$–130 its mean is 0.2501, in agreement with the asymptotic spacing of $1/4 = 2\pi T_{\mathrm{H}} M$.

The behaviour of the real part is not monotonic (panel b). For $\ell = 2$ it decreases from 0.3737 at $n = 0$ to 0.0633 at $n = 9$, next to the unresolved $n = 8$ mode

Table 6.3.: Relative errors (per cent) of the first-order WKB estimate, equation (5.8), and of the eikonal expansion with the $1/L$ correction, equation (4.10), for the overtones $n = 0, 1, 2$, with respect to the exact frequencies from the continued fraction (in units of M^{-1}).

		exact		WKB error		eikonal error	
ℓ	n	$\mathrm{Re}(M\omega)$	$-\mathrm{Im}(M\omega)$	Re	Im	Re	Im
2	0	0.37367	0.08896	+6.7	−0.8	+1.96	+8.2
	1	0.34671	0.27391	+30.8	−14.9	+3.72	+5.4
	2	0.30105	0.47828	+71.7	−28.8	+5.24	+0.6
3	0	0.59944	0.09270	+2.9	−0.4	+0.43	+3.8
	1	0.58264	0.28130	+13.6	−8.3	+0.71	+2.6
	2	0.55168	0.47909	+31.4	−18.1	+0.82	+0.4
4	0	0.80918	0.09416	+1.6	−0.2	+0.15	+2.2
	1	0.79663	0.28433	+8.0	−5.3	+0.24	+1.5
	2	0.77271	0.47991	+18.9	−12.4	+0.26	+0.3

(Section 6.4), then recovers, reaches a maximum of 0.0886 at $n = 16$ and decreases slowly again to 0.0644 at $n = 130$. The limit is $\ln 3/8\pi = 0.043712$ of equation (4.11), so that the $n = 130$ value is still 47 per cent above it. The approach is from above, and I checked that it follows the $n^{-1/2}$ law expected for the leading correction. Writing $\Delta_n = \mathrm{Re}(M\omega_n) - \ln 3/8\pi$, panel (c) shows that $\Delta_n\sqrt{2n+1}$ levels off, taking the values 0.314, 0.324, 0.330 and 0.334 at $n = 40$, 60, 90 and 130. A least-squares fit of $\Delta_n = \gamma_1 x + \gamma_2 x^2 + \gamma_3 x^3$ with $x = (2n+1)^{-1/2}$ over $40 \le n \le 130$ gives $\gamma_1 = 0.340$, which is within 1 per cent of the value 0.343 that Nollert (1993) quotes for $\ell = 2$. The imaginary part is likewise close to its asymptotic form $-(n + \frac{1}{2})/4$, with a difference of 0.019 at $n = 130$ and 0.027 at $n = 60$. These results confirm, with an independent solver, the highly damped behaviour that Nollert (1993) found, and show that the continued fraction with the Nollert remainder remains accurate to overtone numbers well beyond the range in which the frequencies are of any observational interest.

The modes with $\ell = 3$ and 4 behave similarly but are driven towards the imaginary axis more slowly. For $\ell = 3$ the real part falls from 0.5994 to 0.0399 at $n = 38$, the last mode that passes the acceptance criteria, and the sequence then enters the neighbourhood of the algebraically special frequency $-10\mathrm{i}$. For $\ell = 4$ the real part is still 0.1904 at $n = 60$. Higher ℓ therefore delays the approach to the highly damped regime, since the coefficient γ_1 grows with ℓ, as the values quoted in Section 4.5 indicate.

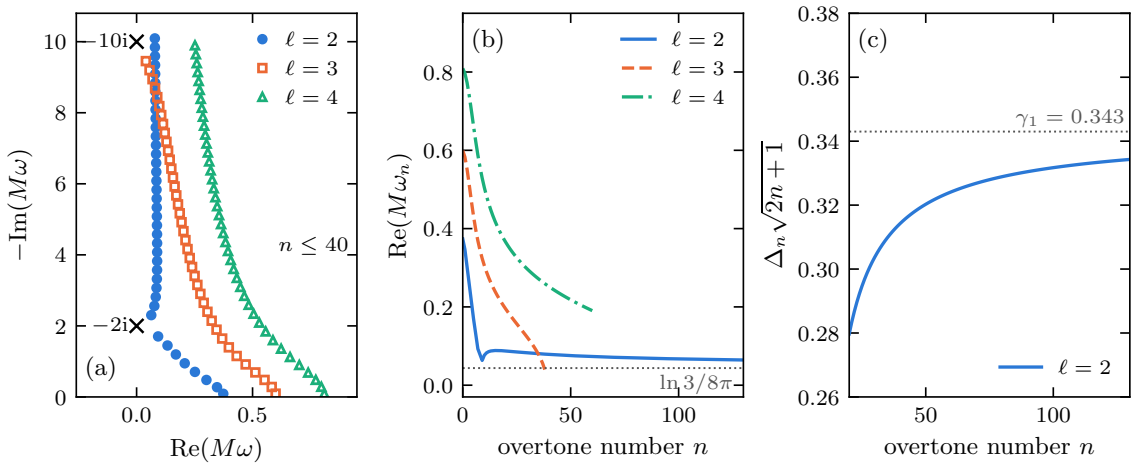


Figure 6.2.: The Schwarzschild gravitational QNM spectrum. (a) Frequencies with $\mathrm{Re}\,\omega > 0$ for $n \le 40$ and $\ell = 2$ (circles), 3 (squares) and 4 (triangles). The crosses mark the algebraically special frequencies −2i and −10i, near which the modes are not resolved (Section 6.4). (b) Real part against overtone number, to $n = 130$ ($\ell = 2$), 38 ($\ell = 3$) and 60 ($\ell = 4$). The dotted line is $\ln 3/8\pi$. (c) The scaled difference $\Delta_n\sqrt{2n+1}$ for $\ell = 2$; the dotted line is the value $\gamma_1 = 0.343$ of Nollert (1993).

6.4. The neighbourhood of the algebraically special frequency

The spectrum has one place where the numerical results are not reliable. For $\ell = 2$ the overtone numbers $n = 7$ and $n = 9$ are at $M\omega = 0.09282 - 1.70384\,\mathrm{i}$ and $0.06326 - 2.30264\,\mathrm{i}$, so that the mode with $n = 8$ lies between them, close to the algebraically special frequency −2i of equation (3.7). The tabulation distributed with the QNM package quotes this mode as $M\omega \approx 1.3 \times 10^{-9} - 2.01352\,\mathrm{i}$ with an uncertainty of 2×10^{-3}, which is between four and seven orders of magnitude larger than that of the neighbouring modes (2×10^{-7} for $n = 9$ and 10^{-10} for $n = 7$), and my own iteration returned $0.0133 - 2.0320\,\mathrm{i}$ with a failed convergence flag, so that the two results do not agree. Solving the problem with different starting values, depths and tolerances did not give a result that satisfied the acceptance criteria. I therefore omit the $\ell = 2$, $n = 8$ mode from the figures and tables. I do not claim that this mode lies exactly at −2i: the tabulated value is 0.0135 away from it, which is larger than the quoted uncertainty, and I have no evidence either way.

The same happens for $\ell = 3$. The modes with $n = 39$ to 43 fail the criteria, and they would lie in the neighbourhood of −10i (the mode $n = 38$ is at $-9.454\,\mathrm{i}$, and my unconverged result for $n = 41$ is at $-10.115\,\mathrm{i}$). For $\ell = 4$ the algebraically special frequency is −30i, which is far beyond the range computed ($\mathrm{Im}\,M\omega = -14.9$ at $n = 60$). The status of the algebraically special modes in the literature is itself

Table 6.4.: Schwarzschild overtones for $\ell = 2$ and 3, in units of M^{-1}. The $\ell = 2$, $n = 8$ mode and the $\ell = 3$ modes with $n \geq 39$ are not resolved (Section 6.4).

n	$\mathrm{Re}(M\omega_n)$	$-\mathrm{Im}(M\omega_n)$
$\ell = 2$		
0	0.373672	0.088962
1	0.346711	0.273915
2	0.301053	0.478277
3	0.251505	0.705148
5	0.169299	1.195608
7	0.092822	1.703841
9	0.063264	2.302645
10	0.076553	2.560827
11	0.082598	2.815442
15	0.088590	3.825105
20	0.087394	5.081467
30	0.082598	7.588683
60	0.073181	15.098232
90	0.068276	22.602642
130	0.064404	32.606046
$\ell = 3$		
0	0.599443	0.092703
1	0.582644	0.281298
2	0.551685	0.479093
3	0.511962	0.690337
7	0.368992	1.643845
10	0.305461	2.397355
20	0.193933	4.918319
30	0.121467	7.433203
38	0.039900	9.453998

unsettled (Kokkotas and Schmidt 1999), and I have not investigated the cause of the numerical difficulty. For the purposes of this dissertation the consequence is that the $\ell = 2$, $n = 8$ mode is not used in any figure or table and that the Kerr overtone ladder of Chapter 7 is restricted to $n \leq 7$, so that its starting values do not depend on this region of the spectrum.

6.5. Time-domain evolution

6.5.1. Ringdown

The time-domain evolution of Section 5.6 gives the response of the black hole to a Gaussian pulse. Figure 6.3(a) shows the signal at $r_* = 20M$ for $\ell = 2$ with time-

Table 6.5.: Fit of a single damped sinusoid to the time-domain signal at $r_* = 20M$ in a window of length $60M$ starting at $t_p + \Delta t$ ($t_p = 27.0M$ for $\ell = 2$ and $21.2M$ for $\ell = 3$ is the time of the largest value of $|\Psi|$ after $t = r_*$). The first two columns give the exact frequency from the continued fraction, the fitted values are ω_R and $\gamma = -\mathrm{Im}\,\omega$, and the last two columns are their relative errors in per cent. The grid spacing is $\Delta r_* = 0.05M$.

ℓ	$\mathrm{Re}(M\omega)$	$-\mathrm{Im}(M\omega)$	window	$M\omega_R$	$M\gamma$	$\Delta\omega_R$	$\Delta\gamma$
2	0.37367	0.08896	$t_p + 20$	0.37164	0.08926	-0.544	$+0.336$
			$t_p + 40$	0.37365	0.08897	-0.006	$+0.009$
			$t_p + 60$	0.37366	0.08894	-0.003	-0.029
3	0.59944	0.09270	$t_p + 20$	0.58970	0.10126	-1.626	$+9.228$
			$t_p + 40$	0.59925	0.09301	-0.032	$+0.332$
			$t_p + 60$	0.59945	0.09271	$+0.001$	$+0.011$

symmetric initial data. The pulse passes the observer between $t \simeq 10M$ and $20M$, and the signal afterwards is a damped oscillation of constant period that has the form expected of the fundamental quasinormal mode. To quantify this, I fitted a function $A\,e^{-\gamma t}\cos(\omega_R t + \varphi)$ with the four parameters free to the signal in a window of length $60M$ beginning at a time Δt after t_p, the time of the largest value of $|\Psi|$ after $t = r_*$. Table 6.5 compares the fitted ω_R and γ with the exact quasinormal frequency from the continued fraction.

For $\ell = 2$, a fit starting $40M$ after the peak recovers the exact frequency to 0.006 per cent in the real part and 0.009 per cent in the damping, and a fit starting $60M$ after it to 0.003 and 0.03 per cent. For $\ell = 3$ the corresponding errors at $40M$ are 0.03 and 0.3 per cent and at $60M$ are 0.001 and 0.01 per cent. Fits that start earlier are worse, with errors of 0.5 per cent in the real part for $\ell = 2$ at $20M$ after the peak, and 1.6 and 9 per cent in ω_R and γ for $\ell = 3$. This is expected, because the first overtone is still present at these times: its damping rate is 0.274 against 0.089 for the fundamental, so that its amplitude relative to the fundamental falls by a factor $e^{-0.185\Delta t}$, which is 0.025 at $\Delta t = 20M$ and 6×10^{-4} at $40M$. The result is a direct demonstration, in a calculation that does not use the quasinormal boundary conditions at all, that the late ringdown is dominated by the fundamental mode of the frequency domain. Running at $\Delta r_* = 0.1M$ changes the fitted values by 2×10^{-5} or less, so the discretisation error is below the precision of the table. It also illustrates the practical difficulty of ringdown spectroscopy: a fit that starts early enough to include much of the signal has to include the overtones, and a fit that starts late enough to ignore them uses a signal that is much smaller.

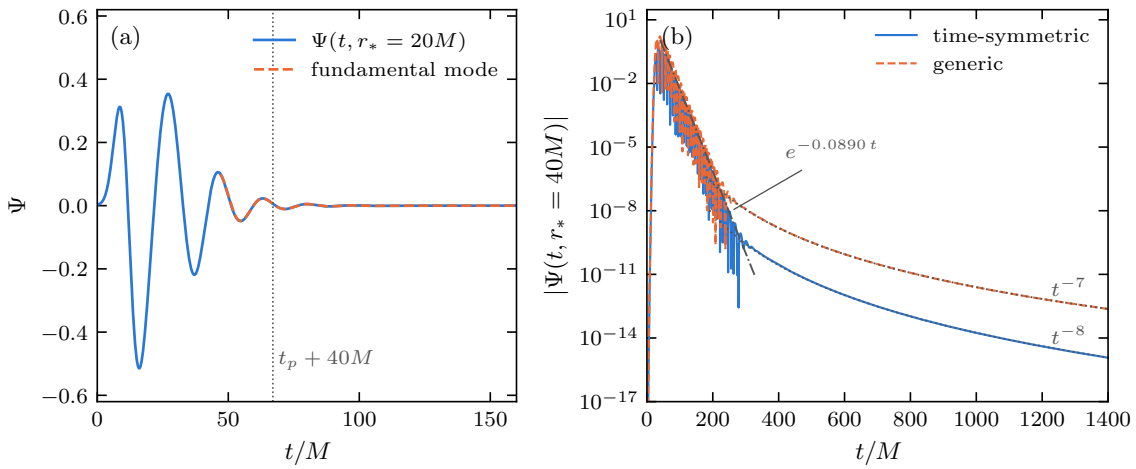


Figure 6.3.: Time-domain response of the Regge–Wheeler equation to a Gaussian pulse (Section 5.6). (a) Signal Ψ at $r_* = 20M$ for $\ell = 2$, time-symmetric data, with the exponentially damped sinusoid of the exact fundamental frequency (dashed), whose amplitude and phase are matched to the data in the window $t_p + 40M \le t \le t_p + 100M$ (the dotted vertical line marks the start of the window). (b) Signal at $r_* = 40M$ for $\ell = 2$ on a logarithmic scale for time-symmetric (solid) and generic (dashed) data. The dotted lines are the power laws t^{-7} and t^{-8} and the dash-dotted line is the exponential decay $e^{-0.0890\,t}$ of the fundamental mode, each normalised to the data as described in the text.

6.5.2. The late-time tail

The same calculation, continued to $t = 1400M$ in extended precision, shows the stage that follows the ringdown. Figure 6.3(b) shows $|\Psi|$ for $\ell = 2$ on a logarithmic scale. The signal first decays exponentially, with the e-folding time of the fundamental mode, and it then departs from this decay and falls as a power of t. To locate the change I fitted an exponential to the peaks of $|\Psi|$ between $40M$ and $140M$ after the pulse, which gives the decay rate 0.0890 of the fundamental mode, and a power law to the signal at $700M < t < 900M$, and took the time at which the two curves cross. For time-symmetric data the crossing is at $t \simeq 284M$ for the observer at $r_* = 40M$ ($292M$ at $r_* = 20M$), where $|\Psi|$ has fallen to 4×10^{-10} (3×10^{-11}) of the amplitude of the initial pulse. For generic data the tail is stronger and the crossing is earlier, at $240M$ ($248M$), when $|\Psi|$ is still 5×10^{-8} (4×10^{-9}). After that time the exponentially damped oscillation is hidden by the tail, which is why ringdown signals can be followed over a limited range only.

The slope of the tail depends on the initial data, as argued in Section 4.3. Table 6.6 gives the local power-law index $p = \mathrm{d}\ln|\Psi|/\mathrm{d}\ln t$ at $t \simeq 600M$ and $800M$, obtained by a straight-line fit of $\ln|\Psi|$ against $\ln t$ over the interval $0.9t < t' < 1.1t$. For generic data the index is -7.03 and -7.02 for $\ell = 2$ and -9.03 and -9.02 for $\ell = 3$, in agreement with the law $t^{-(2\ell+3)}$ of Price (1972), and it is closer to the asymptotic

Table 6.6.: Local power-law index of the late-time tail at $r_* = 40M$, from a straight-line fit of $\ln|\Psi|$ against $\ln t$ over $0.9t < t' < 1.1t$ at $t = 600M$ and $800M$, for the two types of initial data (Section 5.6), with the analytic expectation. Extended-precision runs at $\Delta r_* = 0.2M$.

ℓ	initial data	expected	$p(600M)$	$p(800M)$
2	generic	-7	-7.032	-7.018
2	time-symmetric	-8	-8.055	-8.033
3	generic	-9	-9.033	-9.017
3	time-symmetric	-10	-10.061	-10.083

value at the later time. For time-symmetric data the indices are -8.05 and -8.03 for $\ell = 2$ and -10.06 and -10.08 for $\ell = 3$, which corresponds to $t^{-(2\ell+4)}$, as expected if the leading term proportional to $\partial_t\Psi(0)$ is absent. The $\ell = 2$ values are identical at $\Delta r_* = 0.2M$ and $0.1M$ to three decimal places. For $\ell = 3$ with time-symmetric data the signal has fallen to about 10^{-15} of the initial amplitude by $t = 1000M$, which is the rounding level even of the extended-precision arithmetic, and the fit is therefore only meaningful up to about $t = 800M$. These are exponents measured for a particular discretisation. They agree with the analytic expectation to better than one per cent, and the residual deviation from the asymptotic integer value decreases with time, as it should for a quantity that approaches the asymptotic regime from finite t.

6.6. Summary of the Schwarzschild results

The continued-fraction solver reproduces the tabulated fundamental and low overtone frequencies, and extends them to $n = 130$ for $\ell = 2$. The first-order WKB formula and the eikonal expansion reproduce the fundamental frequency to a few per cent at $\ell = 2$ and much better at higher ℓ, but only the eikonal expansion is accurate for the overtones. The real part of the highly damped modes approaches $\ln 3/8\pi$ from above with the leading correction found by Nollert (1993). The region around the algebraically special frequency could not be resolved. A direct time-domain evolution of the same equation recovers the fundamental frequency to better than 0.03 per cent, in both parts, when the fit starts $60M$ after the peak, and shows the power-law tail, with exponents that match the analytic laws for two types of initial data.

7. Results for Kerr black holes

For a rotating black hole the quasinormal frequencies depend on the spin, and each Schwarzschild point in Figure 6.2 becomes a curve in the complex plane as a/M increases from zero. This chapter follows these curves with the QNM package, as described in Section 5.5, and tests the three patterns that Section 4.6 anticipated: the splitting of the m multiplet, the link with the circular photon orbits and the loss of damping near extremality. A fourth result, about the overtone ladder, was not anticipated. All modes are gravitational ($s = -2$), frequencies are in units of M^{-1}, and the spin a/M is written a.

7.1. The fundamental mode and the overtones as functions of spin

Figure 7.1 shows the two-dimensional version of the diagram of Kokkotas and Schmidt (1999), with the spin as a parameter along each curve. In panel (a) the fundamental modes of the co-rotating series $m = \ell$ with $\ell = 2$–10 start on the Schwarzschild values and move to the right and downwards in the plane of $\mathrm{Re}\,\omega$ against $-\mathrm{Im}\,\omega$, which means higher oscillation frequency and lower damping. In panel (b) the same is shown for the overtones $n = 0$–4 of $\ell = m = 2$. The real part of all of the 45 sequences in this group rises monotonically with spin. The damping decreases slowly at first, and then collapses in the last part of the approach to extremality, so that the curves bend over close to the horizontal axis near their end points.

Figure 7.2 gives the real and imaginary parts for the $\ell = m = 2$ modes with $n = 0$–4, and Table 7.1 lists the fundamental mode. The real part of the fundamental mode rises from 0.3737 at $a = 0$ to 0.6716 at $a = 0.9$ and 0.9385 at $a = 0.998$, a factor of 2.5, while the damping $-\mathrm{Im}(M\omega)$ decreases from 0.0890 to 0.0649 and 0.0145, a factor of 6.1 at $a = 0.998$. The quality factor rises from 2.1 for a Schwarzschild black hole to 5.2 at $a = 0.9$ and 32 at $a = 0.998$. The same remnant therefore rings considerably longer if it is spinning rapidly. For a remnant of the mass and spin that

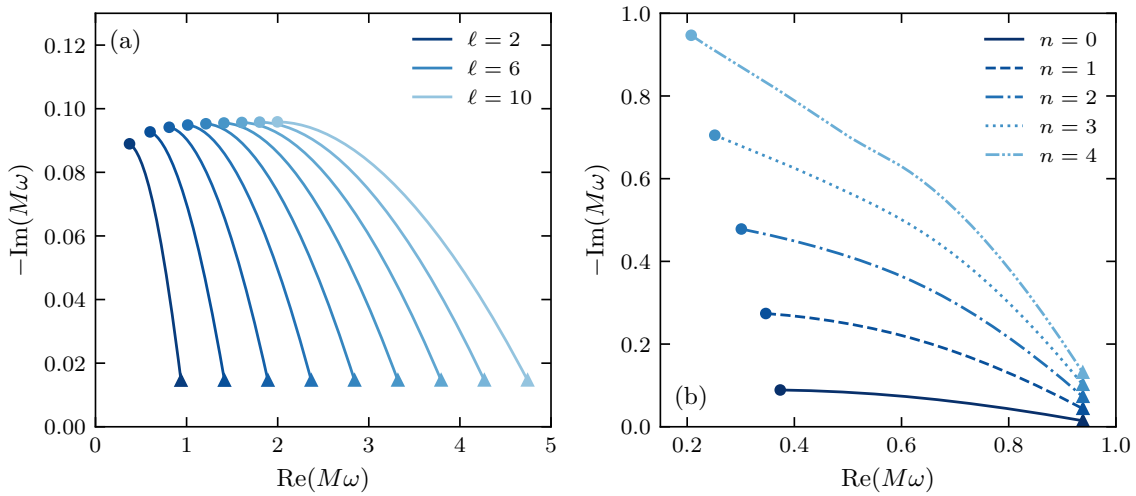


Figure 7.1.: Kerr quasinormal modes as functions of spin, in the plane of $\mathrm{Re}(M\omega)$ and $-\mathrm{Im}(M\omega)$. (a) Fundamental modes $n = 0$, $m = \ell$, for $\ell = 2$–10, each curve running from $a = 0$ (circle) to $a = 0.998$ (triangle). (b) Overtones $n = 0$–4 for $\ell = m = 2$ over the same range of spin.

were inferred for GW150914, $M = 62\,\mathrm{M}_\odot$ and $a = 0.67$ (Abbott et al. 2016), the table gives $f \simeq 271$ Hz, $\tau \simeq 3.7$ ms and $Q \simeq 3.2$. For a $10\,\mathrm{M}_\odot$ hole the fundamental mode at $a = 0.998$ has $f = 3.03$ kHz and $\tau = 3.4$ ms, compared with 1.21 kHz and 0.55 ms at $a = 0$.

The overtones respond differently from the fundamental mode. At $a = 0$ their real parts decrease with n, but as $a \to 1$ they all move towards the same real part, $\mathrm{Re}(M\omega) \to m\Omega_\mathrm{H}M = 1$ for $m = 2$. At $a = 0.998$ the modes $n = 0$–4 have $\mathrm{Re}(M\omega) = 0.9385$, 0.9385, 0.9385, 0.9384 and 0.9382, but they still differ in their damping and form the evenly spaced ladder discussed in Section 7.4.

7.2. Co- and counter-rotating modes and the eikonal limit

Figure 7.3 compares the fundamental modes with $m = +\ell$, which co-rotate with the hole, and $m = -\ell$, which counter-rotate, for $\ell = 2$, 6 and 10. In panel (a) the real part is divided by ℓ. For $\ell \geq 6$ the co-rotating curves are nearly indistinguishable from each other and follow the orbital frequency Ω_pro of the prograde equatorial photon orbit of equation (2.15), while the counter-rotating curves follow Ω_ret. The eikonal result of Section 6.2 therefore carries over to Kerr, with the two circular photon orbits taking the place of the single one. Modes with $m = +\ell$ and $m = -\ell$ correspond to waves that circle the hole with and against its rotation.

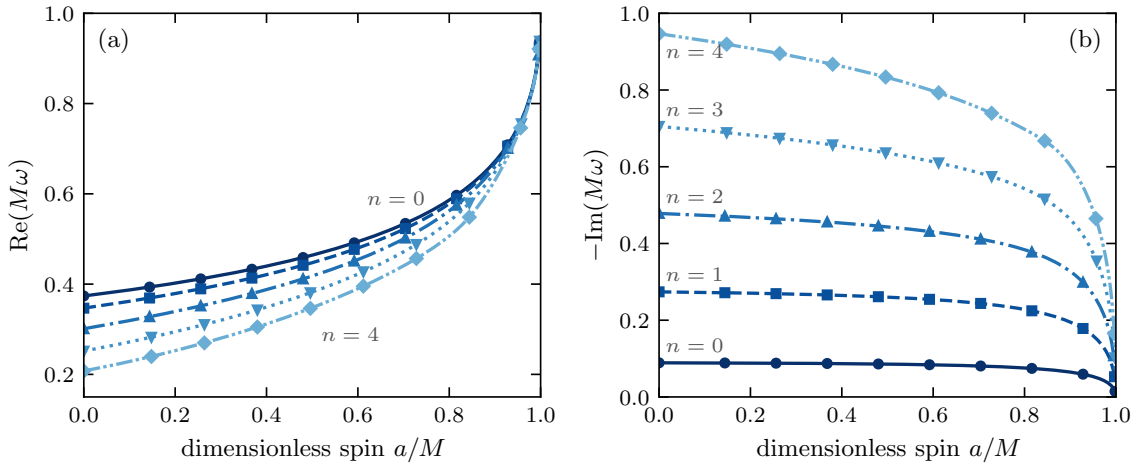


Figure 7.2.: Spin dependence of the $\ell = m = 2$ Kerr frequencies for overtones $n = 0$ (darkest) to 4 (lightest): (a) real part, (b) minus the imaginary part. The markers are placed on a subset of the computed spins.

To quantify this I fitted $\mathrm{Re}(M\omega_{\ell\ell 0}) = \ell\,\Omega + c_0 + c_1/\ell$ at fixed spin over $\ell = 6$–10 and compared the leading coefficient Ω with equation (2.15). The results are in Table 7.2. The fitted frequencies agree with the photon-orbit frequencies to 0.2 per cent or better at every spin tested, for both senses of rotation, and the agreement is somewhat weaker for the co-rotating modes. At $a = 0$ the fit gives 0.1926, against $1/(3\sqrt{3}) = 0.1925$. This is a quantitative confirmation that the large-ℓ behaviour of the real part is set by the null geodesics of the geometry, as proposed by Goebel (1972) and formulated through Lyapunov exponents by Cardoso et al. (2009), and that it holds for rotating black holes with the orbits of Section 2.5.

Panel (b) of Figure 7.3 shows the damping. The counter-rotating modes are barely affected by the spin: for $\ell = 2$ the damping at $a = 0.998$ is 1.0 per cent below its Schwarzschild value, and for $\ell = 10$ it is 3.4 per cent below. The co-rotating modes lose 84–85 per cent of their damping (Table 7.3). For the real parts, panel (c) shows that the fractional change at $a = 0.998$ is nearly independent of ℓ for $\ell \geq 3$, between $+1.34$ and $+1.38$ for the co-rotating modes and between -0.23 and -0.25 for the counter-rotating ones, with $\ell = 2$ ($+1.51$ and -0.22) the only visible exception. In other words, spin more than doubles the oscillation frequency of a co-rotating mode and reduces that of a counter-rotating one by about a quarter, whatever the angular number, and the effect of rotation is close to a multiplicative factor. This is the same asymmetry as the one between the prograde and retrograde photon orbits, whose frequencies at $a = 0.998$ differ by more than a factor of three.

Table 7.1.: Fundamental mode of Kerr black holes ($\ell = m = 2$, $n = 0$) for selected spins, from the fine spin sequence. The frequency f and damping time τ are for a remnant of mass $M = 62\,\mathrm{M}_\odot$, the value inferred for GW150914 (Abbott et al. 2016); the spin $a = 0.67$ is the spin inferred for the same event. $Q = \mathrm{Re}\,\omega/2|\mathrm{Im}\,\omega|$.

a/M	$\mathrm{Re}(M\omega)$	$-\mathrm{Im}(M\omega)$	Q	f (Hz)	τ (ms)
0	0.37367	0.08896	2.10	195	3.43
0.2	0.40215	0.08831	2.28	210	3.46
0.4	0.43984	0.08688	2.53	229	3.51
0.6	0.49404	0.08377	2.95	257	3.65
0.67	0.51986	0.08185	3.18	271	3.73
0.8	0.58602	0.07563	3.87	305	4.04
0.9	0.67161	0.06487	5.18	350	4.71
0.95	0.74632	0.05315	7.02	389	5.75
0.99	0.87089	0.02939	14.82	454	10.39
0.998	0.93852	0.01453	32.29	489	21.01

Table 7.2.: Eikonal frequencies of the Kerr modes. For each spin, Ω_{fit} is the coefficient of ℓ in a fit $\mathrm{Re}(M\omega) = \ell\,\Omega + c_0 + c_1/\ell$ over $\ell = 6$–10 and $n = 0$, for co-rotating ($m = +\ell$) and counter-rotating ($m = -\ell$) modes, compared with the orbital frequency of the prograde (Ω_{pro}) and retrograde (Ω_{ret}) equatorial photon orbit, equation (2.15). Frequencies are in units of M^{-1} and the differences in per cent.

a/M	Ω_{fit} $(m = +\ell)$	Ω_{pro}	diff.	Ω_{fit} $(m = -\ell)$	Ω_{ret}	diff.
0	0.1926	0.1925	+0.08	0.1926	0.1925	+0.08
0.3	0.2193	0.2190	+0.11	0.1733	0.1732	+0.06
0.5	0.2444	0.2441	+0.13	0.1630	0.1629	+0.06
0.7	0.2816	0.2812	+0.16	0.1541	0.1541	+0.05
0.9	0.3522	0.3516	+0.19	0.1464	0.1464	+0.04
0.99	0.4445	0.4441	+0.09	0.1433	0.1432	+0.04

7.3. Rotational splitting of the m multiplet

At $a = 0$ the five modes with $\ell = 2$ and $m = -2, \ldots, 2$ have the same frequency. Figure 7.4 shows how rotation lifts this degeneracy. For small spin the shift is linear in ma, as with the splitting of atomic levels in a magnetic field: at $a = 0.02$ the quantity $[\mathrm{Re}(M\omega_m) - \mathrm{Re}(M\omega_0)]/(m\,a/M)$ is 0.0625, 0.0627, 0.0631 and 0.0633 for $m = -2, -1, +1, +2$ (panel b), so that

$$\mathrm{Re}(M\omega_{2m0}) \simeq 0.3737 + 0.063\,m\,\frac{a}{M}. \tag{7.1}$$

The linear regime is short-lived. At $a = 0.2$ the same ratio ranges from 0.0602 to 0.0676, and at larger spin the splitting is strongly asymmetric between the co-rotating ($m > 0$) and counter-rotating ($m < 0$) modes. At $a = 0.5$ the real parts for

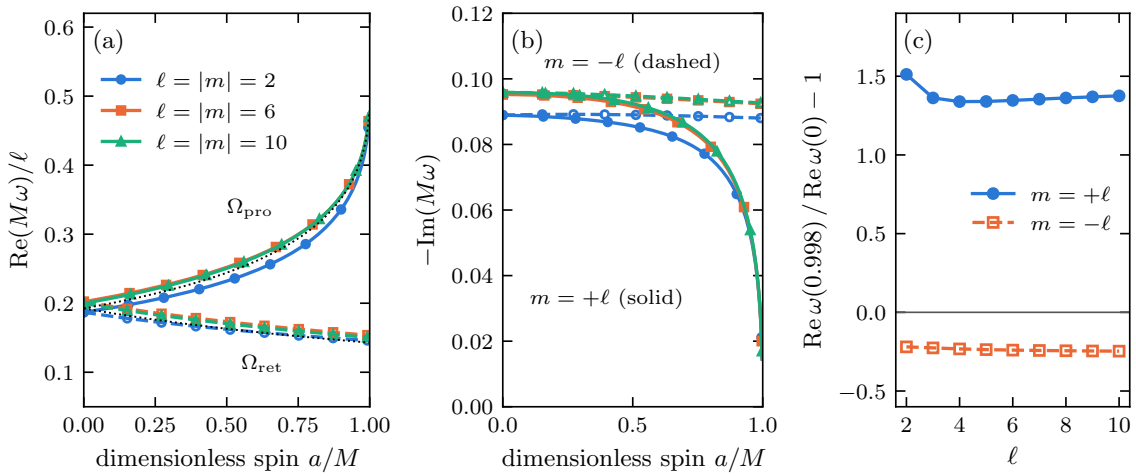


Figure 7.3.: Co-rotating ($m = +\ell$, solid) and counter-rotating ($m = -\ell$, dashed) fundamental modes for $\ell = 2$, 6 and 10. (a) $\mathrm{Re}(M\omega)/\ell$ against spin; the dotted lines are the orbital frequencies $\Omega_{\rm pro}$ and $\Omega_{\rm ret}$ of the equatorial photon orbits. (b) $-\mathrm{Im}(M\omega)$ against spin. (c) Fractional change of the real part between $a = 0$ and $a = 0.998$ for $\ell = 2$–10.

$m = -2, \ldots, 2$ are 0.3243, 0.3515, 0.3833, 0.4206 and 0.4641, and at $a = 0.9$ they are 0.2972, 0.3444, 0.4120, 0.5163 and 0.6716. The $m = 2$ mode has then risen by 80 per cent from its Schwarzschild value, while the $m = -2$ mode has fallen by 20 per cent. The picture of spin as an effective magnetic field is therefore good only for $a \lesssim 0.1$. Beyond this the response is nonlinear and is dominated by the co-rotating modes, which is consistent with the shift $\omega \to \omega - m\Omega_{\rm H}$ of the radial equation of Section 3.7, since $\Omega_{\rm H}$ is itself a nonlinear function of the spin.

7.4. The near-extremal limit

As $a \to M$ the surface gravity (2.12) vanishes, and the co-rotating modes should lose their damping altogether, according to equation (4.12). I tested this by following the $n = 0$ modes of $\ell = m = 2$–10 to $a = 0.99999$, where $\kappa = 2.226 \times 10^{-3}$ (Figure 7.5a and Table 7.4). The ratio $-\mathrm{Im}(M\omega)/(\kappa/2)$ is 0.855 at $a = 0.9$ for $\ell = 2$, increases monotonically with spin, and reaches 0.9984 at $a = 0.99999$, and 0.9987 for $\ell = 10$. At fixed ℓ the deficit from unity is proportional to κ as $\kappa \to 0$. It is $0.709\,\kappa$ for $\ell = 2$, $0.639\,\kappa$ for $\ell = 6$ and $0.576\,\kappa$ for $\ell = 10$ at $a = 0.99999$. Equation (4.12) is then the leading term of an expansion,

$$-\mathrm{Im}(M\omega) = \left(n + \tfrac{1}{2}\right)\kappa\,(1 - c_\ell\kappa + \cdots), \tag{7.2}$$

Table 7.3.: Fractional change of the fundamental-mode frequency between $a = 0$ and $a = 0.998$, $\omega(0.998)/\omega(0) - 1$, for the real and imaginary parts of co-rotating ($m = +\ell$) and counter-rotating ($m = -\ell$) modes.

	$\mathrm{Re}\,\omega$		$\mathrm{Im}\,\omega$	
ℓ	$m = +\ell$	$m = -\ell$	$m = +\ell$	$m = -\ell$
2	+1.512	−0.219	−0.837	−0.010
3	+1.362	−0.226	−0.843	−0.019
4	+1.339	−0.232	−0.846	−0.025
5	+1.340	−0.237	−0.847	−0.029
6	+1.346	−0.240	−0.847	−0.030
7	+1.354	−0.242	−0.847	−0.032
8	+1.362	−0.244	−0.848	−0.033
9	+1.369	−0.246	−0.848	−0.033
10	+1.375	−0.247	−0.848	−0.034

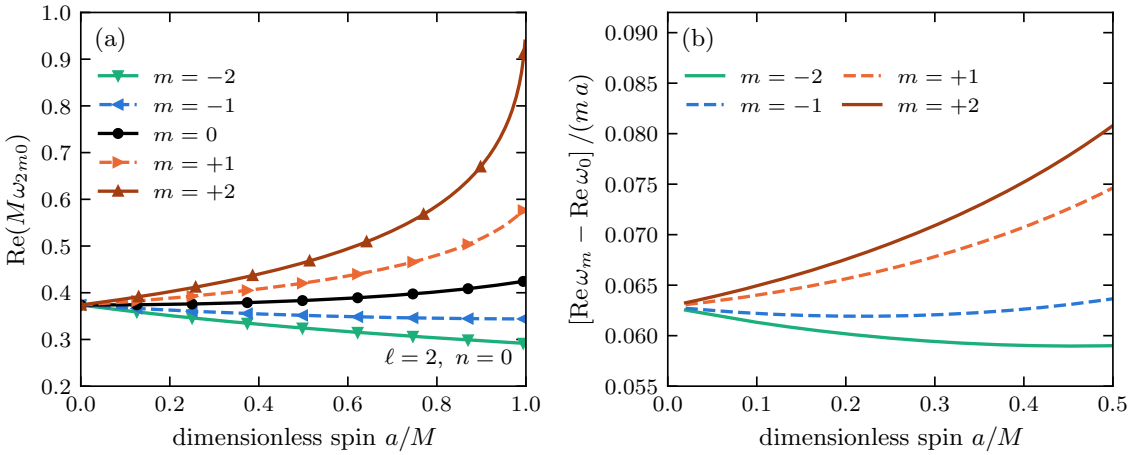


Figure 7.4.: Splitting of the $\ell = 2$, $n = 0$ multiplet by rotation. (a) $\mathrm{Re}(M\omega)$ for $m = -2, \ldots, 2$. (b) The splitting divided by $m\,a/M$ at small spin, showing the linear regime near $a = 0$ with slope ≈ 0.063.

with $c_\ell \simeq 0.6$–0.7 for the fundamental modes. A feature that is striking in the data is that $-\mathrm{Im}(M\omega)$ is the same, 1.111×10^{-3}, to three significant figures for every ℓ from 2 to 10 at $a = 0.99999$, although the real part differs by a factor of five. The damping of these modes is therefore controlled by the near-horizon geometry, through κ, and not by the angular momentum barrier that determines the real part away from extremality.

The overtones that belong to the same family form a ladder (Figure 7.5b). At $a = 0.998$ the $\ell = m = 2$ modes $n = 0, \ldots, 4$ and 6, 7 have $-\mathrm{Im}(M\omega)/[(n' + \frac{1}{2})\kappa] = 0.9778$, 0.9779, 0.9780, 0.9781, 0.9782, 0.9782 and 0.9781, with $n' = n$ for $n \leq 4$ and $n' = n - 1$ for $n = 6, 7$. The spacing of the damping is thus $0.978\,\kappa = 0.0291$, constant to within 0.05 per cent. The mode with $n = 5$ does not belong to this

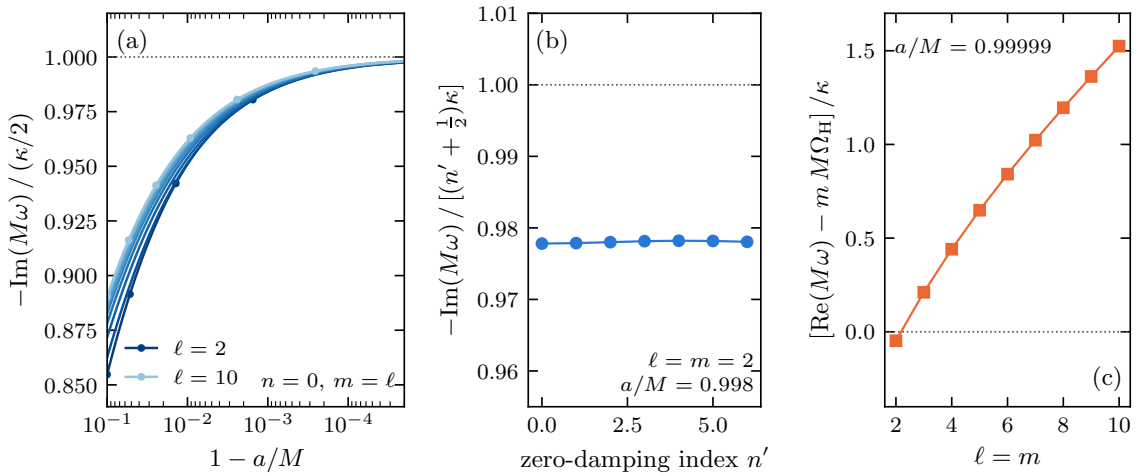


Figure 7.5.: The near-extremal limit. (a) $-\mathrm{Im}(M\omega)/(\kappa/2)$ for the $n = 0$, $m = \ell$ modes with $\ell = 2$ to 10 (darker for lower ℓ) as a function of $1 - a/M$; dots mark computed points of $\ell = 2$ and 10, and the dotted line is the limit of equation (4.12). (b) Damping of the $\ell = m = 2$ overtones at $a = 0.998$ in units of $(n' + \frac{1}{2})\kappa$ (see text for n'); the $n = 5$ mode, which does not lie on the ladder, is not shown. (c) Offset of the real part from $m\Omega_\mathrm{H}$ in units of κ at $a = 0.99999$.

ladder: at $a = 0.998$ it is at $M\omega = 0.5041 - 0.7082\,\mathrm{i}$, far from Re $= 0.9385$. Its damping stays finite, about $24\,\kappa$, as for the damped modes of Yang et al. (2013), and the $n = 6$ and 7 modes are consequently the sixth and seventh zero-damping modes ($n' = 5$, 6).

The real part approaches $m\Omega_\mathrm{H}$, but with a residual that is itself of order κ and depends on ℓ (Figure 7.5c). At $a = 0.99999$ the difference $[\mathrm{Re}(M\omega) - mM\Omega_\mathrm{H}]/\kappa$ rises from -0.048 at $\ell = 2$ to $+1.52$ at $\ell = 10$ (Table 7.4). The real part is therefore below the superradiant threshold of equation (3.17) only for $\ell = 2$, and for $\ell \geq 3$ the modes lie just above it. The modes then are no longer superradiantly amplified in the sense of Section 3.7, and are, in this sense, not modes of the superradiant band, but sit at its edge to within a few units of κ.

7.5. The overtone ladder and avoided crossings

Figure 7.6(a) shows how the $\ell = m = 2$ overtones $n = 0$–7 rearrange as the spin increases. At $a = 0$ they lie along a smooth curve in the complex plane. At $a = 0.5$ the curve is still smooth, but at $a = 0.9$ the $n = 5$ mode, at $0.5369 - 0.7487\,\mathrm{i}$, has a real part below that of both $n = 4$ (0.6298) and $n = 6$ (0.6022), so that it has left the sequence. At $a = 0.998$ the other seven modes have collapsed onto a vertical line at $\mathrm{Re}(M\omega) = 0.938$, as in Section 7.4, while $n = 5$ is isolated at $0.5041 - 0.7082\,\mathrm{i}$.

Figure 7.6(b) follows the three modes $n = 4$, 5 and 6 through the region where

Table 7.4.: Fundamental modes with $m = \ell$ at $a = 0.99999$, where $\kappa = 2.2261 \times 10^{-3}\, M^{-1}$ and $\Omega_{\rm H} = a/(2Mr_+)$ is evaluated exactly. Columns: $\mathrm{Re}(M\omega)$, $-\mathrm{Im}(M\omega)$ in units of 10^{-3}, the ratio $-\mathrm{Im}(M\omega)/(\kappa/2)$, and $[\mathrm{Re}(M\omega) - mM\Omega_{\rm H}]/\kappa$.

ℓ	$\mathrm{Re}(M\omega)$	$-\mathrm{Im}\,(M\omega) \times 10^3$	ratio	$[\mathrm{Re}(M\omega) - mM\Omega_{\rm H}]/\kappa$
2	0.99543	1.1113	0.9984	−0.048
3	1.49378	1.1113	0.9984	+0.211
4	1.99206	1.1113	0.9985	+0.441
5	2.49029	1.1114	0.9985	+0.649
6	2.98849	1.1115	0.9986	+0.841
7	3.48666	1.1115	0.9986	+1.023
8	3.98481	1.1116	0.9987	+1.196
9	4.48295	1.1116	0.9987	+1.363
10	4.98108	1.1116	0.9987	+1.525

this happens. Over most of the range $\mathrm{Re}\,\omega$ of the $n = 5$ mode rises with spin, as for the others. At $a = 0.895$ it reaches a maximum, $0.5380 - 0.7481\,\mathrm{i}$, and then turns back, ending at $0.5041 - 0.7082\,\mathrm{i}$ at $a = 0.998$. In the same range the $n = 4$ and $n = 6$ branches continue towards the ladder at $\mathrm{Re}\,\omega = 0.938$. The closest approach of $n = 5$ to the $n = 6$ branch is $|\Delta\omega| = 0.0667$ at $a = 0.897$, and the distance to the $n = 4$ branch is never smaller than 0.1205 (at $a = 0.8495$). This is the pattern expected for an avoided crossing between two branches, in which the $n = 6$ branch continues smoothly onto the ladder while the $n = 5$ branch is deflected away from it. The branch is a single, smooth curve at every one of its computed points, and the repetitions of the calculation with different resolution of Section 5.5 reproduce it, so that it is not an artefact of mode tracking. I have not examined the eigenfunctions, and so I cannot say whether the transition is a true avoided crossing or a close passage of two separate branches.

The consequence is that the label n, which is assigned by the ordering at $a = 0$, ceases to be a smooth and spin-independent way of identifying a mode once the spin is large. The mode that one would call the sixth overtone at $a = 0.998$, in the order of damping, is not the mode that is connected to $n = 5$ at $a = 0$. Statements about "the nth overtone" of a rapidly rotating hole therefore need the spin at which n is defined. I did not compare this particular crossing with the literature, and I do not claim that it is a new result. The identification of the $n = 5$ branch with the damped family of Yang et al. (2013) rests only on the finite value of its damping.

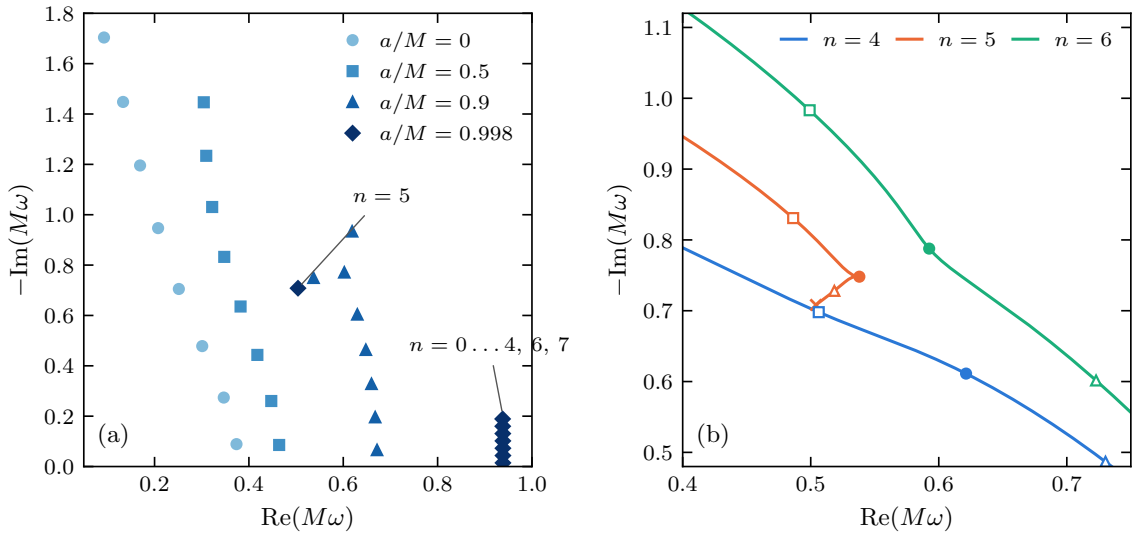


Figure 7.6.: Overtone ladder of the $\ell = m = 2$ modes. (a) Frequencies for $n = 0$–7 at four spins. At $a = 0.998$ all modes except $n = 5$ lie at $\mathrm{Re}(M\omega) \simeq 0.938$. (b) Trajectories of $n = 4$, 5 and 6 in the complex plane from $a = 0$ (upper left, outside the frame) to $a = 0.998$ (crosses). Markers show the position at $a = 0.8$ (squares), 0.895 (circles) and 0.95 (triangles).

7.6. Summary of the Kerr results

The scans show that rotation does three things to the Schwarzschild spectrum. First, it speeds up and de-damps the co-rotating modes while leaving the counter-rotating modes almost unchanged, with fractional changes that are nearly independent of ℓ for $\ell \geq 3$. Second, it makes the large-ℓ real part follow the orbital frequency of the prograde or retrograde photon orbit, to 0.2 per cent. Third, close to extremality it brings the co-rotating modes to the superradiant edge $m\Omega_{\mathrm{H}}$ with damping $\kappa(n' + \frac{1}{2})$, and it reorganises the overtone ladder so that the $n = 5$ mode of $\ell = m = 2$ departs from the others after a close approach to its neighbour.

8. Discussion

This chapter collects what Chapters 6 and 7 show, discusses what the results do and do not imply for the measurement of ringdown signals, and sets out the limits of the calculations.

8.1. The role of the light ring

The clearest result of the computations is that the real part of the frequency at large ℓ is set by null geodesics. For a Schwarzschild black hole the eikonal expansion with the $1/L$ correction reproduces the real part of the fundamental mode to 2.0 per cent at $\ell = 2$ and to 0.15 per cent at $\ell = 4$ (Section 6.2). For Kerr black holes, the coefficient of ℓ in the fitted real part agrees with the orbital frequency of the prograde or retrograde equatorial photon orbit to 0.2 per cent at every spin tested (Table 7.2). The asymmetry between co- and counter-rotating modes in Section 7.2 follows from the same fact: at $a = 0.998$ the orbital frequencies of the two photon orbits differ by more than a factor of three, and the fundamental frequencies of the two senses of rotation change by about $+135$ per cent and -24 per cent respectively, with little dependence on ℓ. It is not necessary to solve the perturbation equation to anticipate the sign and rough size of the effect of spin on the real part.

For the damping the situation is less complete. In the Schwarzschild case the imaginary part at leading order is $-(n + \frac{1}{2})$ times the Lyapunov exponent $\lambda_{\mathrm{L}} = 1/3\sqrt{3}M$ of the light ring, and this holds to 8 per cent at $\ell = 2$ and 0.4 per cent at $\ell = 10$. For Kerr the damping of the co-rotating modes near extremality is instead controlled by the surface gravity, $-\mathrm{Im}(M\omega) \simeq 0.978\,(n' + \frac{1}{2})\kappa$ for the ladder at $a = 0.998$, and at $a = 0.99999$ it is independent of ℓ to three figures (Section 7.4). The two descriptions are not in conflict, because they meet at extremality: the radius of the prograde photon orbit tends to the horizon radius M, and its orbital frequency tends to $\Omega_{\mathrm{H}} = 1/2M$ (Section 2.5), so the eikonal real part $\ell\Omega_{\mathrm{pro}}$ and the horizon value $m\Omega_{\mathrm{H}}$ coincide for $m = \ell$. What the data show is that at $a = 0.99999$ the real part lies within about $1.5\,\kappa$ of $m\Omega_{\mathrm{H}}$ for all $\ell \leq 10$ (Table 7.4). I did not

compute the Lyapunov exponents of the Kerr photon orbits, and so I cannot say at which spin, if any, the light-ring description of the damping is replaced by the horizon one. That these modes accumulate at the edge of the frequency band in which a wave can extract rotational energy, $\omega < m\Omega_{\rm H}$ (Section 3.7), is a statement about their frequencies only. The amplification factor of a wave at the quasinormal frequency has not been computed.

8.2. Consequences for ringdown spectroscopy

The quality factor of the fundamental $\ell = m = 2$ mode rises from 2.1 at $a = 0$ to 5.2 at $a = 0.9$ and 32 at $a = 0.998$ (Section 7.1). For a remnant like that of GW150914 the mode completes about one oscillation per e-folding time ($f \simeq 271$ Hz, $\tau \simeq 3.7$ ms), so that very little of the signal is oscillatory. The spin of the remnant therefore decides how many cycles of the fundamental mode are available to a fit, and a rapidly spinning remnant is a much better source for spectroscopy than a slowly spinning one.

The time-domain experiment of Section 6.5 illustrates the second difficulty. In a noise-free signal a single damped sinusoid recovers the fundamental frequency to 0.01 per cent for $\ell = 2$ and to 0.3 per cent in the damping for $\ell = 3$ if it is fitted from $40M$ after the peak, but by then the amplitude has fallen to $e^{-0.0890\times 40} \simeq 0.03$ of its value at the peak. A fit that begins earlier has to include the first overtone, which decays three times as fast (0.274 against 0.089 for $\ell = 2$), so that its amplitude relative to the fundamental falls by a factor of 0.025 in $20M$. The amplitude of the overtones relative to the fundamental in a real merger is not determined by the frequencies, and I have not computed it. Whether an overtone can be measured, and when the linear description applies at all after the merger, are questions that these calculations do not address.

The Kerr overtones add a different complication. At $a = 0.998$ the overtones $n = 0$–4 of $\ell = m = 2$ have the same real part to 3×10^{-4} and differ only in damping, which is $(n + \frac{1}{2}) \times 0.0291$, so that the first overtone damps three times as fast as the fundamental. A fit of a rapidly spinning remnant can therefore separate these modes only through their damping times. The additional result of Section 7.5, that the label n is defined at $a = 0$ and does not follow the damping order at high spin, means that a statement about "the nth overtone" of a particular remnant needs the spin at which n is defined.

8.3. Stability of the overtones

The frequencies in this dissertation are those of the exact vacuum potentials. Section 4.7 discussed the stability of the black holes themselves. A different question is how the *spectrum* responds when the potential is changed slightly, for example by matter around the hole, and here the overtones behave differently from the fundamental mode. Jaramillo et al. (2021) analysed this with the pseudospectrum and found that the overtones are spectrally unstable: a small perturbation of the potential with a short length scale can move them by an amount that is much larger than its size. Cheung et al. (2022) then showed that a small perturbation placed far from the black hole can displace even the fundamental frequency considerably. The consequences for the observed signal are still being worked out. The highly damped modes of Section 6.3, which I computed to $n = 130$ to test the solver, are in this sense mathematical properties of the idealised potential and not observable quantities. The same caution applies to the overtones of the Kerr ladder.

8.4. Limitations

The results are subject to the following restrictions, some of which were stated in Chapter 1.

- The calculation is linear, in vacuum, and for gravitational perturbations ($s = -2$) only. I did not compute the quasinormal modes of scalar, electromagnetic or higher-order perturbations. Excitation factors, and hence the amplitudes of the modes, are not computed.

- The Schwarzschild computation uses the Regge–Wheeler equation. By isospectrality (Section 3.3) the frequencies are those of the Zerilli equation as well, but I did not solve the Zerilli equation directly, and so the numerical results do not test that equivalence.

- The modes $\ell = 2$, $n = 8$ and $\ell = 3$, $n = 39$–43 could not be resolved (Section 6.4), and the Kerr overtone ladder stops at $n = 7$. The Kerr scans cover $\ell \leq 10$ and overtones $n \leq 4$ ($n \leq 7$ for $\ell = m = 2$), and in the analysis reach $a = 0.99999$ for the fundamental modes and $a = 0.998$ for everything else.

- The Kerr frequencies come from an existing package and not from a solver of my own. I checked them against my Schwarzschild solver at $a = 0$ and against the photon-orbit frequencies and the surface-gravity law, and I tested

the tracking of the overtones (Section 5.5), but I did not repeat the Kerr calculation with an independent method.

- The $n = 5$ branch of Section 7.5 is identified from the frequencies alone. I did not compute eigenfunctions, so I cannot say whether the transition is an avoided crossing in the strict sense.

- The tail exponents of Section 6.5 are measured with one discretisation of one initial condition, at two resolutions. They agree with the analytic laws to better than 1 per cent, but they do not constitute an independent derivation of those laws.

9. Conclusions and future work

9.1. Conclusions

This dissertation set out to understand the quasinormal modes of Schwarzschild and Kerr black holes, from the geometry of the solutions to the spectrum, and to test numerically some of the physical pictures that are used to interpret it. The main conclusions are the following.

1. The Regge–Wheeler equation follows from the linearised Einstein equations for odd-parity perturbations, and I verified the derivation symbolically (Appendix A). The Regge–Wheeler and Zerilli potentials are related by the supersymmetric partner construction, which I also verified symbolically, and the position of the potential maximum is expanded about the light ring in powers of $1/\ell(\ell+1)$ (Chapter 3).

2. An independent implementation of Leaver's continued fraction, with the Nollert remainder, reproduces the tabulated Schwarzschild frequencies for $n \leq 20$ to 7×10^{-7} at $\ell = 2$ and to 2×10^{-10} at $\ell = 3$ and 4, and extends the $\ell = 2$ overtones to $n = 130$. The real part of the highly damped modes approaches $\ln 3/8\pi$ from above with the correction $\gamma_1(2n+1)^{-1/2}$, with $\gamma_1 = 0.340$, against the value 0.343 of Nollert (1993), and the spacing of the imaginary parts approaches $1/4M$ (Chapter 6).

3. The first-order WKB formula is accurate for the fundamental mode, to 6.7 per cent at $\ell = 2$ and 0.28 per cent at $\ell = 10$, but not for the overtones, for which the error at $\ell = 2$ reaches 31 per cent at $n = 1$ and 72 per cent at $n = 2$. The eikonal expansion with the $1/L$ correction is the better analytic estimate, with errors in the real part of at most 5.2 per cent at $\ell = 2$ and 0.9 per cent at $\ell = 3$ and 4 up to $n = 2$.

4. A direct time-domain integration of the Regge–Wheeler equation recovers the fundamental quasinormal frequency, in both the real and imaginary parts, to

0.03 per cent or better when the fit begins $60M$ after the peak. It shows the late-time power-law tail, with exponents -7.0 and -9.0 for generic data and -8.0 and -10.1 for time-symmetric data at $\ell = 2$ and 3, in agreement with the laws $t^{-(2\ell+3)}$ and $t^{-(2\ell+4)}$.

5. The Kerr fundamental modes with $m = +\ell$ and $m = -\ell$ behave in opposite ways. At $a = 0.998$ the real part of the co-rotating mode is increased by a factor of between 2.3 and 2.4 for $\ell \geq 3$ and its damping is reduced by 84–85 per cent, whereas for the counter-rotating mode the real part falls by about a quarter and the damping is nearly unchanged. At large ℓ the real part follows ℓ times the orbital frequency of the prograde or retrograde equatorial photon orbit to 0.2 per cent (Chapter 7).

6. As $a \to M$ the damping of the co-rotating modes follows $\frac{1}{2}\kappa$ for the fundamental mode, with a deficit proportional to κ whose coefficient is 0.58–0.71 for $\ell = 2$–10, and it is independent of ℓ to three figures at $a = 0.99999$. The $\ell = m = 2$ overtones $n = 0$–4, 6 and 7 form a ladder with spacing $0.978\,\kappa$, and their real parts converge to $m\Omega_{\mathrm{H}}$.

7. The $n = 5$ overtone of $\ell = m = 2$ leaves this ladder at high spin, after its closest approach of 0.0667 to the $n = 6$ branch at $a = 0.897$, so that the overtone label defined at $a = 0$ does not follow the order of damping at large spin. I identified this from the frequencies alone.

8. The region around the algebraically special frequency could not be resolved: the $\ell = 2$, $n = 8$ and the $\ell = 3$, $n = 39$–43 modes fail the acceptance criteria.

Items 2 to 4 reproduce results that are known in the literature with a solver and a time-domain code of my own, and they agree with them. Items 5 and 6 confirm the expectations of Section 4.6, the link with the photon orbits and the zero-damping modes, and the numerical values quoted are those of the scans designed for this dissertation. Item 7 I have not compared with the literature, and I do not claim that it is new.

9.2. Future work

The limitations of Section 8.4 suggest several extensions.

- *Excitation factors.* The frequencies determine which modes exist but not how strongly an initial perturbation excites them. The excitation factors of the

Schwarzschild and Kerr modes, and their dependence on n and on the spin, are needed to decide which of the overtones of Chapter 7 could be present in a ringdown signal.

- *The algebraically special region.* A separate treatment of the neighbourhood of $\omega = -2\mathrm{i}$ for $\ell = 2$ and of $-10\mathrm{i}$ for $\ell = 3$, for example with a different expansion of the radial function, would resolve the $n = 8$ mode and show whether the Schwarzschild spectrum has modes at the algebraically special frequencies.

- *The avoided crossing.* The eigenfunctions and the excitation factors of the $n = 5$ branch would show whether the transition of Section 7.5 is an avoided crossing. A scan over other (ℓ, m) would show whether similar rearrangements happen elsewhere in the spectrum.

- *The Kerr damping and the light ring.* A calculation of the Lyapunov exponent of the equatorial photon orbits in the Kerr geometry, compared with the damping computed here, would test the geometric interpretation of the imaginary part for rotating black holes, which I tested for the real part only.

- *Independent Kerr calculations.* A time-domain evolution of the Teukolsky equation, or a separate frequency-domain solver, would provide the independent check of the Kerr results that is available for the Schwarzschild ones.

- *Environments and spectral instability.* The effect of a small perturbation of the potential on the overtones, in the pseudospectrum framework mentioned in Section 8.3, could be explored with the same solvers, with a modified potential.

Bibliography

B. P. Abbott et al. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116:061102, 2016.

J. M. Bardeen, W. H. Press, and S. A. Teukolsky. Rotating black holes: Locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *ApJ*, 178:347, 1972.

E. Berti, V. Cardoso, and A. O. Starinets. Quasinormal modes of black holes and black branes. *Classical and Quantum Gravity*, 26:163001, 2009.

R. Brito, V. Cardoso, and P. Pani. *Superradiance*, volume 906 of *Lecture Notes in Physics*. Springer, 2015.

V. Cardoso, A. S. Miranda, E. Berti, H. Witek, and V. T. Zanchin. Geodesic stability, Lyapunov exponents, and quasinormal modes. *Physical Review D*, 79: 064016, 2009.

V. Cardoso, S. Biswas, and S. Sarkar. The physics of black holes and their environments: Consequences for gravitational wave science. arXiv e-prints, 2025.

S. Chandrasekhar. *The Mathematical Theory of Black Holes*. Oxford University Press, 1983.

S. Chandrasekhar. On algebraically special perturbations of black holes. *Proceedings of the Royal Society of London Series A*, 392:1, 1984.

S. Chandrasekhar and S. Detweiler. The quasi-normal modes of the Schwarzschild black hole. *Proceedings of the Royal Society of London Series A*, 344:441, 1975.

M. H.-Y. Cheung, K. Destounis, R. Panosso Macedo, E. Berti, and V. Cardoso. Destabilizing the fundamental mode of black holes: The elephant and the flea. *Physical Review Letters*, 128:111103, 2022.

D. Christodoulou. Reversible and irreversible transformations in black-hole physics. *Physical Review Letters*, 25:1596, 1970.

G. B. Cook and M. Zalutskiy. Gravitational perturbations of the Kerr geometry: High-accuracy study. *Physical Review D*, 90:124021, 2014.

M. Dafermos, G. Holzegel, and I. Rodnianski. The linear stability of the Schwarzschild solution to gravitational perturbations. *Acta Mathematica*, 222: 1–214, 2019.

S. Detweiler. Black holes and gravitational waves. III. the resonant frequencies of rotating holes. *ApJ*, 239:292, 1980.

S. R. Dolan and A. C. Ottewill. On an expansion method for black hole quasinormal modes and Regge poles. *Classical and Quantum Gravity*, 26:225003, 2009.

V. Ferrari and B. Mashhoon. New approach to the quasinormal modes of a black hole. *Physical Review D*, 30:295, 1984.

C. J. Goebel. Comments on the "vibrations" of a black hole. *ApJ*, 172:L95, 1972.

D. Häfner, P. Hintz, and A. Vasy. Linear stability of slowly rotating Kerr black holes. *Inventiones Mathematicae*, 223:1227, 2021.

S. Hod. Bohr's correspondence principle and the area spectrum of quantum black holes. *Physical Review Letters*, 81:4293, 1998.

S. Hod. Slow relaxation of rapidly rotating black holes. *Physical Review D*, 78: 084035, 2008.

M. Isi, M. Giesler, W. M. Farr, M. A. Scheel, and S. A. Teukolsky. Testing the no-hair theorem with GW150914. *Physical Review Letters*, 123:111102, 2019.

S. Iyer and C. M. Will. Black-hole normal modes: A WKB approach. I. foundations and application of a higher-order WKB analysis of potential-barrier scattering. *Physical Review D*, 35:3621, 1987.

J. L. Jaramillo, R. Panosso Macedo, and L. Al Sheikh. Pseudospectrum and black hole quasinormal mode instability. *Physical Review X*, 11:031003, 2021.

R. P. Kerr. Gravitational field of a spinning mass as an example of algebraically special metrics. *Physical Review Letters*, 11:237, 1963.

K. D. Kokkotas and B. G. Schmidt. Quasi-normal modes of stars and black holes. *Living Reviews in Relativity*, 2:2, 1999.

E. W. Leaver. An analytic representation for the quasi-normal modes of Kerr black holes. *Proceedings of the Royal Society of London Series A*, 402:285, 1985.

A. Maassen van den Brink. Analytic treatment of black-hole gravitational waves at the algebraically special frequency. *Physical Review D*, 62:064009, 2000.

C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. W. H. Freeman, San Francisco, 1973.

L. Motl and A. Neitzke. Asymptotic black hole quasinormal frequencies. *Advances in Theoretical and Mathematical Physics*, 7:307, 2003.

H.-P. Nollert. Quasinormal modes of Schwarzschild black holes: The determination of quasinormal frequencies with very large imaginary parts. *Physical Review D*, 47:5253, 1993.

H.-P. Nollert. Quasinormal modes: the characteristic 'sound' of black holes and neutron stars. *Classical and Quantum Gravity*, 16:R159, 1999.

R. Penrose. Gravitational collapse: the role of general relativity. *Rivista del Nuovo Cimento*, 1:252, 1969.

W. H. Press. Long wave trains of gravitational waves from a vibrating black hole. *ApJ*, 170:L105, 1971.

W. H. Press and S. A. Teukolsky. Floating orbits, superradiant scattering and the black-hole bomb. *Nature*, 238:211, 1972.

R. H. Price. Nonspherical perturbations of relativistic gravitational collapse. I. scalar and gravitational perturbations. *Physical Review D*, 5:2419, 1972.

T. Regge and J. A. Wheeler. Stability of a Schwarzschild singularity. *Physical Review*, 108:1063, 1957.

B. F. Schutz and C. M. Will. Black hole normal modes - a semianalytic approach. *ApJ*, 291:L33, 1985.

Y. Shlapentokh-Rothman. Quantitative mode stability for the wave equation on the Kerr spacetime. *Communications in Mathematical Physics*, 329:859, 2015.

L. C. Stein. qnm: A Python package for calculating Kerr quasinormal modes, separation constants, and spherical-spheroidal mixing coefficients. *Journal of Open Source Software*, 4:1683, 2019.

S. A. Teukolsky. Perturbations of a rotating black hole. I. fundamental equations for gravitational, electromagnetic, and neutrino-field perturbations. *ApJ*, 185:635, 1973.

C. V. Vishveshwara. Scattering of gravitational radiation by a Schwarzschild black-hole. *Nature*, 227:936, 1970.

B. F. Whiting. Mode stability of the Kerr black hole. *Journal of Mathematical Physics*, 30:1301, 1989.

H. Yang, F. Zhang, A. Zimmerman, D. A. Nichols, E. Berti, and Y. Chen. Branching of quasinormal modes for nearly extremal Kerr black holes. *Physical Review D*, 87:041502, 2013.

F. J. Zerilli. Effective potential for even-parity Regge-Wheeler gravitational perturbation equations. *Physical Review Letters*, 24:737, 1970.

A. Derivation of the Regge–Wheeler equation

Spherical symmetry allows one to take $m = 0$ without loss of generality. Odd-parity perturbations of the Schwarzschild metric (2.1), in the Regge–Wheeler gauge, have only two non-zero components,

$$\begin{aligned} h_{t\phi} &= h_0(t,r)\sin\theta\,\partial_\theta P_\ell(\cos\theta), \\ h_{r\phi} &= h_1(t,r)\sin\theta\,\partial_\theta P_\ell(\cos\theta). \end{aligned} \tag{A.1}$$

I linearised the Ricci tensor about the background by computing the Christoffel symbols of $g + \epsilon h$ to first order in ϵ with a computer-algebra system, and reduced the angular derivatives with Legendre's equation (the script is `derive_rw.py`, Appendix D). As a check of the code, a pure-gauge perturbation $h_{ab} = \nabla_a \xi_b + \nabla_b \xi_a$, with ξ_a having only an r component, gives $\delta R_{ab} = 0$ identically. For the odd-parity ansatz the only non-vanishing components of δR_{ab} are $t\phi$, $r\phi$ and $\theta\phi$, and setting them to zero gives

$$f\left(\partial_r^2 h_0 - \partial_t\partial_r h_1 - \frac{2}{r}\partial_t h_1\right) - \left[\frac{\ell(\ell+1)}{r^2} - \frac{4M}{r^3}\right] h_0 = 0, \tag{A.2}$$

$$\frac{1}{f}\left(\partial_t^2 h_1 - \partial_t\partial_r h_0 + \frac{2}{r}\partial_t h_0\right) + \frac{(\ell-1)(\ell+2)}{r^2}\, h_1 = 0, \tag{A.3}$$

$$\frac{1}{f}\,\partial_t h_0 - \partial_r\left(f h_1\right) = 0. \tag{A.4}$$

With the time dependence $e^{-\mathrm{i}\omega t}$, $h_0 = H_0(r)e^{-\mathrm{i}\omega t}$ and $h_1 = H_1(r)e^{-\mathrm{i}\omega t}$, equation (A.4) gives H_0 algebraically in terms of H_1,

$$H_0 = \frac{\mathrm{i}f}{\omega}\left(f H_1\right)'. \tag{A.5}$$

Substituting into equation (A.3) and writing $X = fH_1$ gives

$$\frac{\omega^2}{f}X + (fX')' - \frac{2f}{r}X' - \frac{(\ell-1)(\ell+2)}{r^2}X = 0. \tag{A.6}$$

The substitution $X = r\Psi$ removes the first-derivative term, and using $f' = 2M/r^2$ the result is

$$f\,(f\Psi')' + \left(\omega^2 - f\left[\frac{(\ell-1)(\ell+2)}{r^2} + \frac{2f}{r^2} - \frac{f'}{r}\right]\right)\Psi = 0. \tag{A.7}$$

Since $(\ell-1)(\ell+2) = \ell(\ell+1)-2$ and $2f/r^2 - f'/r = 2/r^2 - 6M/r^3$, the bracket equals $\ell(\ell+1)/r^2 - 6M/r^3$. This is the Regge–Wheeler equation, $\mathrm{d}^2\Psi/\mathrm{d}r_*^2 + (\omega^2 - V_{\mathrm{RW}})\Psi = 0$, with V_{RW} of equation (3.3) and $\Psi = fh_1/r$. Equation (A.2) is not independent: its consistency with the other two follows from the Bianchi identity. I confirmed with the computer-algebra system that equation (A.3), with H_0 eliminated, equals the Regge–Wheeler operator acting on Ψ up to the overall factor $-r^3/[2(r-2M)^2]$, which does not depend on Ψ.

B. Derivation of Leaver's recurrence relation

This appendix derives the recurrence relation (5.3) of Section 5.1. In units $2M = 1$, with $L = \ell(\ell+1)$, $f = 1-1/r$ and $x = (r-1)/r$, so that $r = 1/(1-x)$, the equation to solve is

$$\mathcal{E}[\Psi] \equiv f\,\frac{\mathrm{d}}{\mathrm{d}r}\left(f\frac{\mathrm{d}\Psi}{\mathrm{d}r}\right) + \left(\omega^2 - V\right)\Psi = 0, \qquad V = f\left[\frac{L}{r^2} + \frac{1-s^2}{r^3}\right]. \tag{B.1}$$

Write the solution as $\Psi = P(r)\sum_n a_n x^n$ with the prefactor $P(r) = (r-1)^{-\mathrm{i}\omega} r^{2\mathrm{i}\omega} e^{\mathrm{i}\omega(r-1)}$ of equation (5.2). Acting with $\mathcal{E}$ on a single term $P\,x^n$ and dividing by $P\,x^n$ gives a polynomial in x of degree four,

$$\frac{\mathcal{E}[P\,x^n]}{P\,x^n} = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4, \tag{B.2}$$

with the coefficients

$$\begin{aligned}
c_0 &= n(n - 2\mathrm{i}\omega),\\
c_1 &= -L - 4n^2 + 12\mathrm{i}n\omega - 2n + 8\omega^2 + 4\mathrm{i}\omega + s^2 - 1,\\
c_2 &= 2L + 6n^2 - 22\mathrm{i}n\omega + 6n - 20\omega^2 - 12\mathrm{i}\omega - 3s^2 + 3,\\
c_3 &= -L - 4n^2 + 16\mathrm{i}n\omega - 6n + 16\omega^2 + 12\mathrm{i}\omega + 3s^2 - 3,\\
c_4 &= -(-n + 2\mathrm{i}\omega + s - 1)(n - 2\mathrm{i}\omega + s + 1).
\end{aligned}$$

These were obtained with a computer-algebra system (Appendix D). Dividing the polynomial by $(1-x)^2$ leaves no remainder, so that

$$c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 = (1-x)^2\left[q_0(n) + q_1(n)\,x + q_2(n)\,x^2\right], \tag{B.3}$$

with

$$
\begin{aligned}
q_0(n) &= n(n - 2\mathrm{i}\omega), \\
q_1(n) &= -L - 2n^2 + 8\mathrm{i}n\omega - 2n + 8\omega^2 + 4\mathrm{i}\omega + s^2 - 1, \\
q_2(n) &= (n + 1 - 2\mathrm{i}\omega)^2 - s^2.
\end{aligned}
$$

The factor $(1 - x)^2$ is common to all terms and can be dropped, and the condition $\mathcal{E}[\Psi] = 0$ becomes

$$\sum_n a_n \, x^n \left[q_0(n) + q_1(n) \, x + q_2(n) \, x^2 \right] = 0. \tag{B.4}$$

The coefficient of x^{n+1} vanishes if

$$q_0(n + 1) \, a_{n+1} + q_1(n) \, a_n + q_2(n - 1) \, a_{n-1} = 0. \tag{B.5}$$

With $\rho = -\mathrm{i}\omega$ and $\varepsilon = s^2 - 1$ this is the recurrence (5.3), with $\alpha_n = q_0(n + 1)$, $\beta_n = q_1(n)$ and $\gamma_n = q_2(n - 1)$ as in equation (5.4); the script confirms that the two forms of the three coefficients are identical. Since $q_0(0) = 0$ and $a_{-1} = 0$, the first equation of the series is $\alpha_0 a_1 + \beta_0 a_0 = 0$.

C. Asymptotic series for the continued-fraction remainder

For large k the ratio $R_k = a_{k+1}/a_k$ of successive coefficients of the recurrence (5.3) satisfies $\alpha_k R_k + \beta_k + \gamma_k/R_{k-1} = 0$. Insert the series

$$R_k = 1 + \frac{C_1}{\sqrt{k}} + \frac{C_2}{k} + \frac{C_3}{k^{3/2}} + \frac{C_4}{k^2} + O\big(k^{-5/2}\big)\,, \tag{C.1}$$

expand R_{k-1} with $k^{-1/2} \to (k-1)^{-1/2} = k^{-1/2}(1-1/k)^{-1/2}$, and equate the coefficients of successive powers of $k^{-1/2}$. The leading orders cancel identically, the first condition is $C_1^2 = 2\rho$, and each later order fixes the next coefficient. With $L = \ell(\ell+1)$ and $\varepsilon = s^2 - 1$,

$$C_1 = \pm\sqrt{2\rho}, \qquad C_2 = 2\rho - \tfrac{3}{4}, \tag{C.2}$$

$$C_3 = \pm\frac{\sqrt{2}\,[16L + 16\rho(4\rho - 5) + 3]}{64\sqrt{\rho}}, \tag{C.3}$$

$$C_4 = \frac{16L + 16\rho\,(4L - 4\varepsilon - 16\rho + 5) + 3}{128\rho}. \tag{C.4}$$

The signs of C_1 and C_3 are the same. The sign is chosen to give the minimal solution, that is the ratio of smaller modulus. For $\rho = -\mathrm{i}\omega$ the first two coefficients are $C_1 = \pm\sqrt{-2\mathrm{i}\omega}$ and $C_2 = -(3/4 + 2\mathrm{i}\omega)$, in agreement with Nollert (1993). The coefficients C_3 and C_4 extend his result. The derivation was done symbolically (`verify_nollert.py`), which also confirms the closed forms above. As a numerical test, for $\ell = 2$, $M\omega = 0.373672 - 0.088962\,\mathrm{i}$ and $k = 10^3$, the series differs from the ratio found by iterating the recurrence downwards from $k = 2 \times 10^5$ by 9.4×10^{-8}, against 8.6×10^{-5} if only C_1 and C_2 are retained.

D. Program listings

D.1. Scripts

All computations were done with the scripts listed in Table D.1. They are run from the directory `code/` of the source archive, read their input from `data/` and write their output to `data/`, `tables/` and `figs/`. The two programs that carry the main numerical results are listed in full in the sections that follow.

Table D.1.: Scripts used for this dissertation.

Script	Purpose
`derive_rw.py`	Linearised Ricci tensor for odd-parity perturbations, gauge check, reduction to the Regge–Wheeler equation (Appendix A).
`verify_susy.py`	Superpartner relation between the Regge–Wheeler and Zerilli potentials, algebraically special frequency, series for the maximum of the potential (Chapter 3).
`verify_leaver_recurrence.py`	Derivation of the recurrence relation (Appendix B).
`verify_nollert.py`	Asymptotic series for the remainder of the continued fraction (Appendix C).
`verify_potential.py`	Maximum of the potential and first-order WKB frequencies (Table 6.1).
`leaver_schw.py`, `leaver_fast.py`	Leaver solver for Schwarzschild in multiple and in double precision.
`compute_schw.py`	Schwarzschild overtones to high order, with the acceptance criteria of Section 5.2.
`exact_extra.py`	Fundamental frequencies for $\ell = 2$–5, 10, 20 and 100.
`convergence.py`	Convergence of the continued fraction with depth (Table 5.1).
`wkb_overtones.py`	WKB and eikonal estimates for the overtones (Table 6.3).
`compute_kerr.py`, `test_branch.py`	Kerr spin sequences with the QNM package, and the repetition of the $n = 4$–6 sequences at higher resolution (Section 5.5).
`timedomain.py`, `td_price.py`, `td_analyze.py`	Time-domain evolution in double and in extended precision, and the analysis of the ringdown (Section 6.5).
`gen_tables_diss.py`, `gen_tables_new.py`	Tables of Chapters 5–7.
`make_figs.py`	Figures of Chapters 2, 6 and 7.
`final_checks.py`, `verify_text.py`	Comparison of the solver with the tabulation of the QNM package, and a check of the numbers quoted in the text.

D.2. Leaver solver for Schwarzschild

The routine below is the double-precision solver of Sections 5.1 and 5.2. The function `residual` evaluates the left-hand side of equation (5.6) for a trial frequency, with the remainder of Appendix C at the cutoff, and `solve` finds its root by the secant method.

Listing D.1: `leaver_fast.py`: continued-fraction solver (double precision).

```
"""
Double-precision Schwarzschild Leaver solver (the multiprecision version is leaver_schw.py
and the derivation of the tail is in verify_nollert.py).  Units 2M = 1 internally; results are M*
    omega.
"""
import cmath
import numpy as np

# Coefficients of the remainder series, derived symbolically in verify_nollert.py
def _c(rho, L, eps):
    sr = cmath.sqrt(rho)
    c1 = cmath.sqrt(2) * sr
    c2 = 2 * rho - 0.75
    c3 = cmath.sqrt(2) * (16 * L + 16 * rho * (4 * rho - 5) + 3) / (64 * sr)
    c4 = (16 * L + 16 * rho * (4 * L - 4 * eps - 16 * rho + 5) + 3) / (128 * rho)
    return c1, c2, c3, c4


def _tail(N, rho, L, eps):
    c1, c2, c3, c4 = _c(rho, L, eps)
    x = N ** -0.5
    best = None
    for s in (1.0, -1.0):
        r = 1 + s * c1 * x + c2 * x**2 + s * c3 * x**3 + c4 * x**4
        if best is None or abs(r) < abs(best):
            best = r
    return best


def residual(w2M, l, n, s=2, N=3000):
    rho = -1j * w2M
    eps = s * s - 1
    L = l * (l + 1)
    r = _tail(N, rho, L, eps)
    for k in range(N, n, -1):
        al = k * k + (2 * rho + 2) * k + 2 * rho + 1
        be = -(2 * k * k + (8 * rho + 2) * k + 8 * rho**2 + 4 * rho + L - eps)
        ga = k * k + 4 * rho * k + 4 * rho**2 - eps - 1
        r = -ga / (be + al * r)
    def al(k): return k * k + (2 * rho + 2) * k + 2 * rho + 1
    def be(k): return -(2 * k * k + (8 * rho + 2) * k + 8 * rho**2 + 4 * rho + L - eps)
    def ga(k): return k * k + 4 * rho * k + 4 * rho**2 - eps - 1
    if n == 0:
        return be(0) + al(0) * r
    u = -be(0) / al(0)
```

```
    for k in range(1, n):
        u = -(be(k) + ga(k) / u) / al(k)
    return be(n) + ga(n) / u + al(n) * r


def solve(l, n, guess_M, s=2, N=3000, tol=1e-13, maxit=60):
    """Secant iteration on the n-th inversion; guess and result are M*omega."""
    f = lambda w: residual(w, l, n, s=s, N=N)
    w0 = 2 * complex(guess_M)
    w1 = w0 * (1 + 1e-5) + 1e-6
    f0, f1 = f(w0), f(w1)
    for _ in range(maxit):
        if f1 == f0:
            break
        w2 = w1 - f1 * (w1 - w0) / (f1 - f0)
        w0, f0 = w1, f1
        w1, f1 = w2, f(w2)
        if abs(w1 - w0) < tol * max(1.0, abs(w1)):
            return w1 / 2, True
    return w1 / 2, False
```

D.3. Time-domain evolution

This is the program used for the ringdown study of Section 6.5. The extended-precision version used for the tails differs only in the floating-point type, the size of the domain and the choice of initial data.

Listing D.2: `timedomain.py`: leapfrog evolution of the Regge–Wheeler equation.

```
"""
Time-domain evolution of the Regge-Wheeler equation (M = 1, s = 2, odd parity).

    d^2 Psi/dt^2 = d^2 Psi/dr*^2 - V(r) Psi,        V = f [ l(l+1)/r^2 - 6/r^3 ]

Explicit second-order leapfrog on a uniform r* grid.  Initial data: a Gaussian
Psi(0, r*) = exp[-(r*-r0)^2/(2 sigma^2)],  dPsi/dt(0) = 0.  The signal is recorded
at fixed r*_obs.  Run at two resolutions to estimate the discretisation error.
The domain is large enough that no boundary reflection reaches the observer
before the end of the run (checked by the choice of L1, L2, tmax below).
Output: ../data/timedomain.pkl
"""
import pickle
import sys
import time
import numpy as np
from scipy.special import lambertw


def r_of_rstar(x):
    """Invert r* = r + 2 ln(r/2 - 1) (M = 1) for r > 2, vectorised, stable for large |r*|."""
    y = x / 2.0 - 1.0                       # r*/2 - 1 = r/2 + ln(r/2 - 1) - 1 ; with u = r/2 - 1:
      u + ln u = y
    # solve u + ln u = y  =>  u = W(exp(y))
    u = np.empty_like(y)
    small = y < 600
    u[small] = lambertw(np.exp(y[small])).real
    w = np.maximum(y[~small], 1.0)
    for _ in range(60):                     # Newton on w + ln w - y = 0
        w = w - (w + np.log(w) - y[~small]) / (1 + 1 / w)
    u[~small] = w
    return 2.0 * (u + 1.0)


def run(l, dx, L1=-600.0, L2=1000.0, tmax=900.0, cfl=0.5, r0=10.0, sigma=3.0, robs=(20.0, 40.0,
     100.0)):
    x = np.arange(L1, L2 + 0.5 * dx, dx)
    r = r_of_rstar(x)
    f = 1 - 2 / r
    V = f * (l * (l + 1) / r**2 - 6 / r**3)
    dt = cfl * dx
    nsteps = int(round(tmax / dt))
    psi0 = np.exp(-((x - r0) ** 2) / (2 * sigma**2))
    lap = np.zeros_like(psi0)
    lap[1:-1] = (psi0[2:] - 2 * psi0[1:-1] + psi0[:-2]) / dx**2
    psi1 = psi0 + 0.5 * dt**2 * (lap - V * psi0)          # second-order start, dPsi/dt(0) = 0
    idx = [int(round((ro - L1) / dx)) for ro in robs]
```

```
    t = [0.0, dt]
    sig = [[psi0[i], psi1[i]] for i in idx]
    pm, p = psi0, psi1
    c2 = (dt / dx) ** 2
    for k in range(2, nsteps + 1):
        pn = np.empty_like(p)
        pn[1:-1] = 2 * p[1:-1] - pm[1:-1] + c2 * (p[2:] - 2 * p[1:-1] + p[:-2]) - dt**2 * V[1:-1]
      * p[1:-1]
        pn[0] = pn[-1] = 0.0
        pm, p = p, pn
        t.append(k * dt)
        for j, i in enumerate(idx):
            sig[j].append(p[i])
    return np.array(t), {ro: np.array(s) for ro, s in zip(robs, sig)}


if __name__ == "__main__":
    out = {}
    for l in (2, 3):
        for dx in (0.1, 0.05):
            t0 = time.time()
            t, s = run(l, dx)
            print(f"l={l} dx={dx}: {len(t)} steps, {time.time() - t0:.1f}s", flush=True)
            out[(l, dx)] = (t, s)
    pickle.dump(out, open("../data/timedomain.pkl", "wb"))
```